

Home Lab Project Log

Windows Server 2022 | Active Directory | DNS | Group Policy | Microsoft 365

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A record of tasks completed while building a home lab using Windows Server 2022, Active Directory, Group Policy and ServiceNow. The goal was to get hands-on experience with the tools used in IT support and platform engineering roles.

01

Configure Static IP Address on Windows Server

Networking | TCP/IPv4

Set a static IP address on the server so it always has the same address on the network. This is required before promoting it to a Domain Controller.

Steps

- Pressed Windows + R and typed `ncpa.cpl` then pressed Enter to open Network Connections
- Right-clicked the Ethernet adapter and selected Properties
- Double-clicked Internet Protocol Version 4 (TCP/IPv4)
- Selected Use the following IP address and filled in the details below
- Selected Use the following DNS server addresses and entered the DNS settings
- Clicked OK then closed the Ethernet Properties window

Settings Applied

- **IP Address:** 192.168.1.60
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1
- **Preferred DNS:** 192.168.1.60 (points to itself as it will run DNS)
- **Alternate DNS:** 8.8.8.8

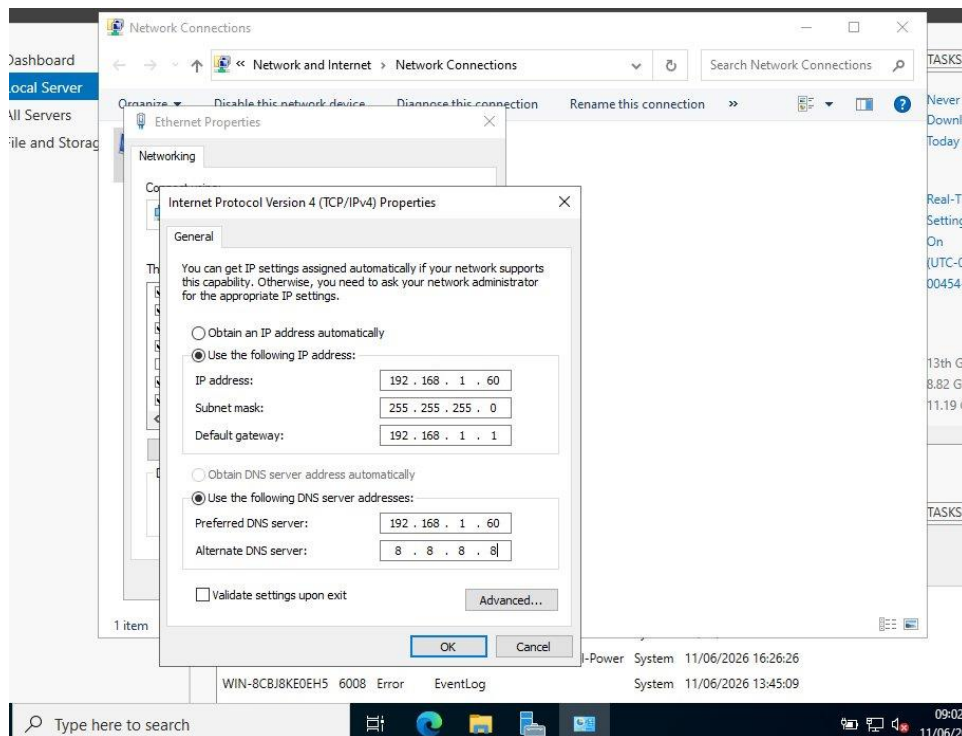


Fig 1 - Static IPv4 address configured on Windows Server 2022

Why This Matters

A Domain Controller needs a static IP so other machines can always find it. If the IP keeps changing, domain joins and DNS lookups will fail. The server points to itself for DNS because it will be running the DNS role for the domain.

02

Install AD DS and DNS, Promote Server to Domain Controller

Active Directory | DNS | Server Roles

Used Server Manager to install the AD DS and DNS roles, then promoted the server to a Domain Controller for a new domain called mylab.local.

Steps

- Opened Server Manager and clicked Manage then Add Roles and Features
- Selected Role-based or feature-based installation and clicked Next
- Selected the local server and clicked Next
- Ticked Active Directory Domain Services and DNS Server from the roles list
- Clicked Add Features when prompted then clicked Next through to Install
- After installation completed, clicked the notification flag and selected Promote this server to a domain controller
- Selected Add a new forest and entered the root domain name: mylab.local
- Set the Directory Services Restore Mode (DSRM) password and clicked Next through the wizard

- Clicked Install and waited for the server to restart automatically
- Server rebooted and logged back in as MYLAB\Administrator confirming the promotion was successful

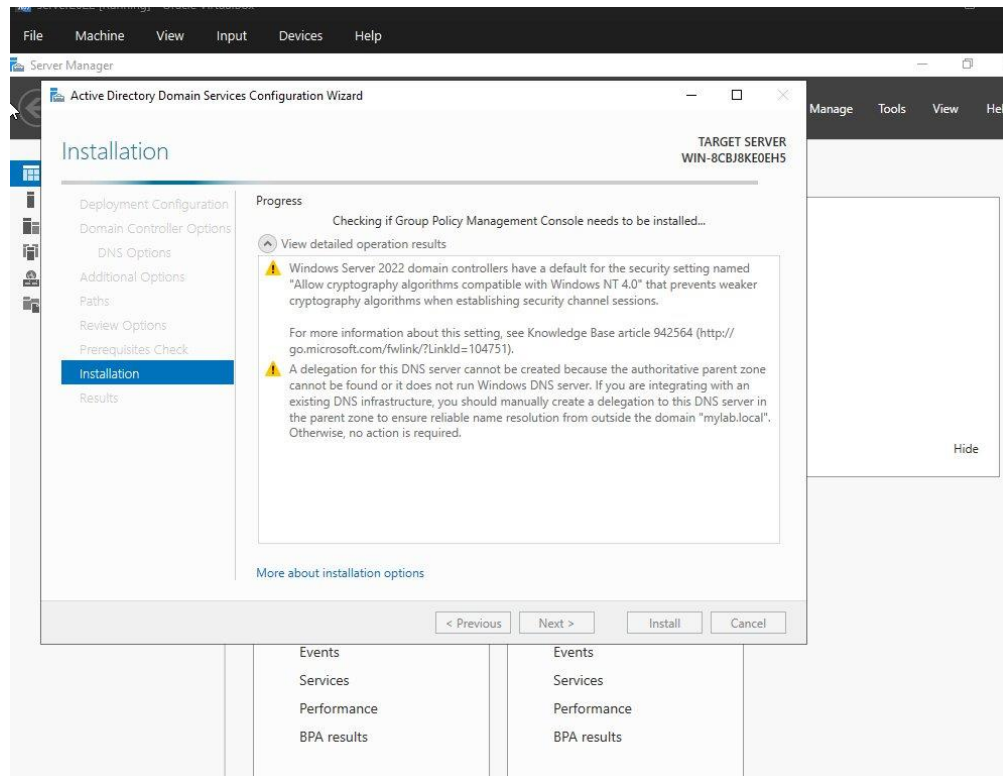


Fig 2 - AD DS Configuration Wizard during installation

Why This Matters

Promoting the server to a Domain Controller is the foundation of the whole lab. AD DS manages authentication for all users and computers, and DNS makes sure internal names like mylab.local resolve correctly.

03

Join a Windows 10 PC to the Domain

Active Directory | Domain Join

Joined a Windows 10 machine to mylab.local and verified it appeared in Active Directory Users and Computers on the server.

Steps on the Windows 10 Machine

- Right-clicked the Start button and selected System
- Clicked Rename this PC (advanced) to open System Properties
- Clicked Change under Computer Name/Domain Changes
- Selected Domain and typed mylab.local then clicked OK
- Entered Domain Admin username and password when prompted

- Clicked OK on the welcome message and restarted the machine

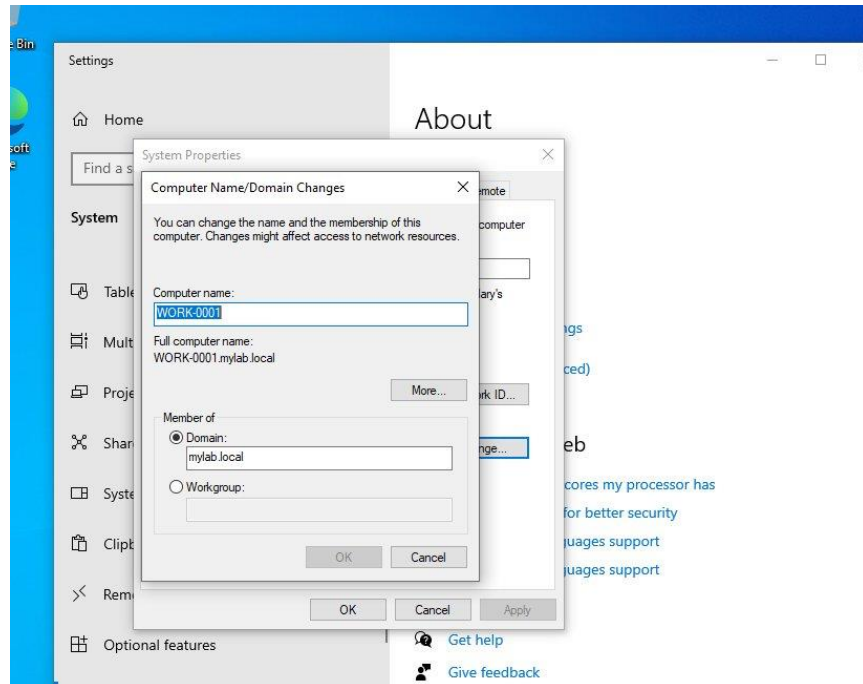


Fig 3 - Windows 10 System Properties showing WORK-001 joined to mylab.local

Verified on the Server

- Opened Active Directory Users and Computers (ADUC) on the Domain Controller
- Expanded mylab.local and clicked the Computers container
- WORK-001 appeared in the list confirming the domain join was successful

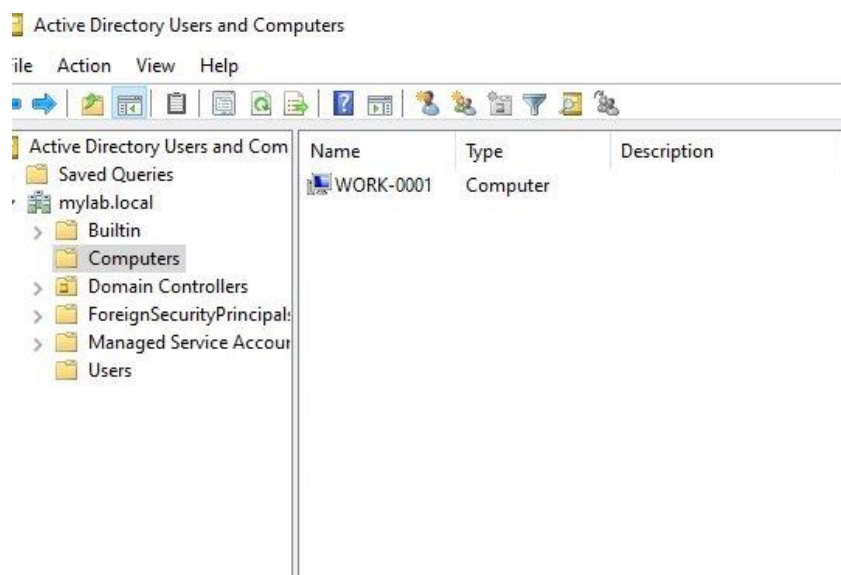


Fig 4 - ADUC showing WORK-001 in the Computers container

Why This Matters

Once a PC is joined to the domain it is managed centrally. Group Policies and user permissions set in AD apply to it automatically, and any domain user can log in without needing a local account on that machine.

04

Create an Organisational Unit and User Account in Active Directory

Active Directory | User Management

Created an Organisational Unit (OU) for HR and then added a new user account inside it. OUs are used to organise users and computers in AD so policies can be targeted at specific departments.

Steps to Create the OU

- Opened Active Directory Users and Computers (ADUC) from Server Manager or by typing dsa.msc in the Run box
- Right-clicked on mylab.local in the left panel
- Selected New then Organisational Unit
- Typed HR as the name and clicked OK
- The HR OU appeared under mylab.local in the tree

Steps to Create the User

- Right-clicked the HR OU and selected New then User
- Entered First name: Priya, Last name: Nair, Full name: Priya Nair
- Set the User logon name to pnair and the domain to mylab.local then clicked Next
- Set a temporary password and ticked User must change password at next logon
- Clicked Next then Finish to create the account

The screenshot shows the 'New Object - User' dialog box in the Active Directory Users and Computers console. The 'Create in' field is set to 'mylab.local/Domain Controllers/HR'. The user details are as follows:

Field	Value
First name:	Priya
Last name:	Nair
Full name:	Priya Nair
User logon name:	pnair@mylab.local
User logon name (pre-Windows 2000):	MYLAB\pnair

The 'Next >' button is highlighted, indicating the next step in the process.

Fig 5 - Creating user Priya Nair in the HR OU

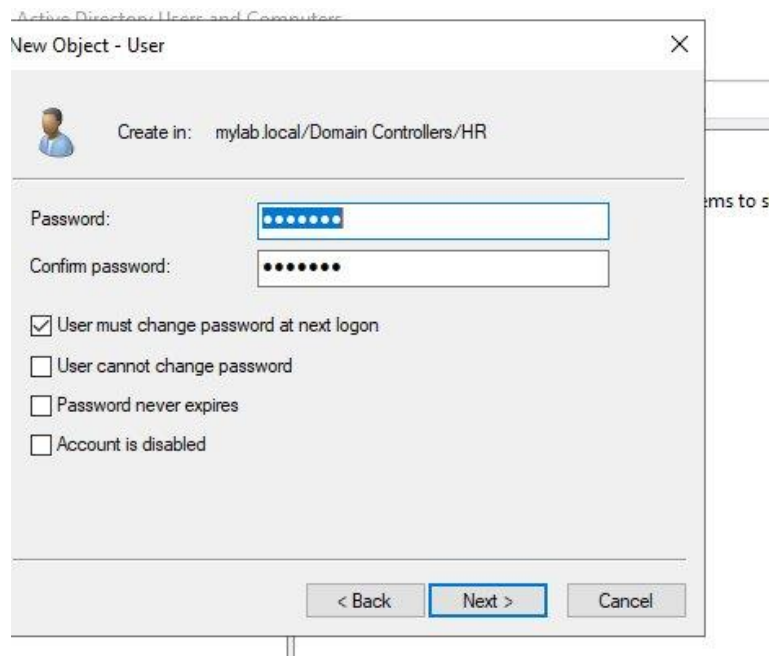


Fig 6 - Password settings applied to the new user account

Why This Matters

Placing the account in the HR OU means any HR Group Policies apply to it automatically. Forcing a password change at first login means the admin never knows the user's actual password.

05

Create a Security Group and Assign Admin Privileges

Active Directory | Security Groups

Created a security group called tech support, added it to Domain Admins to give it elevated privileges, then added the user cobu man to the group so they inherited admin rights through membership.

Steps to Create the OU and Group

- Opened ADUC and right-clicked mylab.local
- Selected New then Organisational Unit and named it Tech Support
- Right-clicked the Tech Support OU and selected New then Group
- Named the group tech support, set Group scope to Global and Group type to Security then clicked OK

Steps to Assign Admin Privileges to the Group

- Right-clicked the tech support group and selected Properties
- Clicked the Member Of tab then clicked Add
- Typed Domain Admins and clicked OK to add the group to Domain Admins

Steps to Create the User and Add to Group

- Right-clicked the Tech Support OU and selected New then User

- Created the user account for cobu man
- Right-clicked the tech support group and selected Properties
- Clicked the Members tab then clicked Add
- Typed cobu man and clicked OK to add them to the group

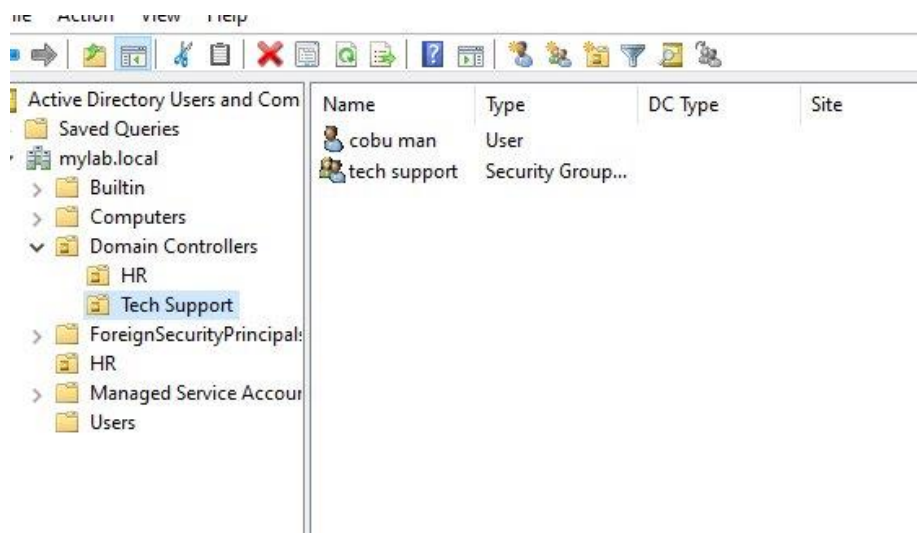


Fig 7 - Tech Support OU showing cobu man and the tech support security group

Why This Matters

Admin rights should always be assigned to groups, not individual users. If cobu man leaves, removing them from the group instantly removes their privileges without touching anything else.

06

Set Up Tiered Support Groups Using Least Privilege

Active Directory | Security Groups | Least Privilege

Built a tiered IT support structure with three security groups inside the Tech Support OU, each representing a different level of access. A user was created and assigned to each tier.

Steps to Create the Groups

- Opened ADUC and expanded the Tech Support OU
- Right-clicked the Tech Support OU and created two sub-OUs named Groups and Users to keep things organised
- Right-clicked the Groups OU and selected New then Group
- Created 1st line support with Group scope Global and type Security, clicked OK
- Repeated the same steps to create 2nd line support and 3rd line group

Steps to Create Users and Assign to Groups

- Right-clicked the Users OU and selected New then User
- Created John Smith and set his logon name, then right-clicked the 1st line support group, selected Properties, Members tab, and added John Smith

- Created Sarah Jones the same way and added her to 2nd line support
- Created Marcus Log and added him to 3rd line group

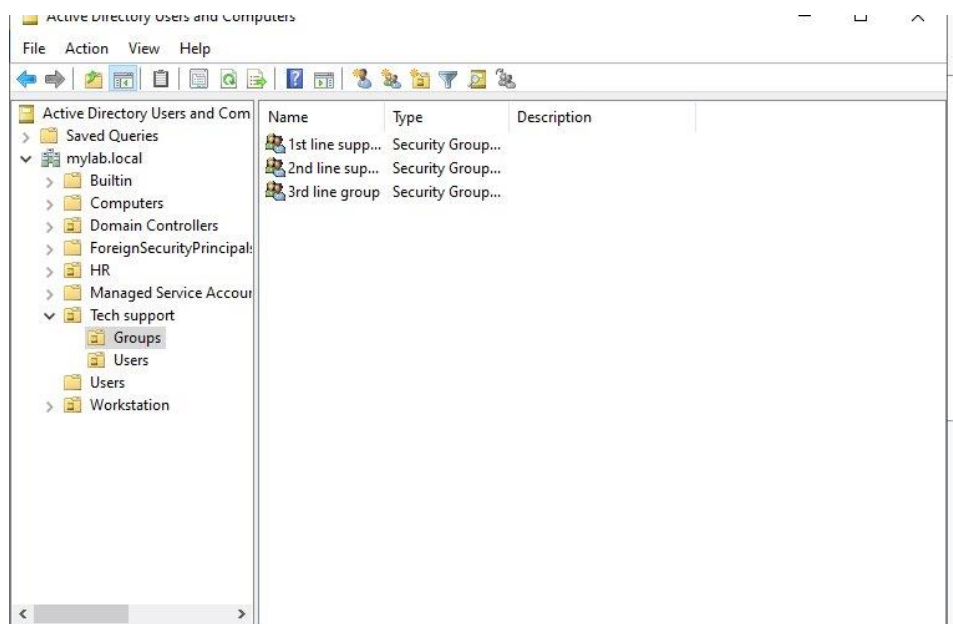


Fig 8 - Groups OU showing the three tiered security groups

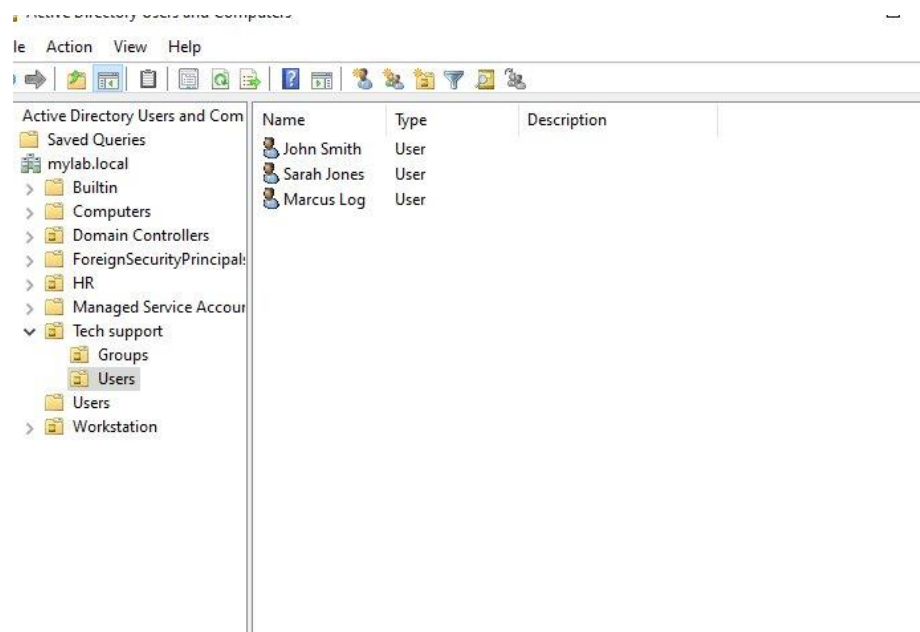


Fig 9 - Users OU showing the three support staff accounts

Why This Matters

Least privilege means users only get the access they need for their role. A 1st line agent does not need the same permissions as a senior engineer. Using groups makes this easy to manage as people change roles.

A user called to say they could not log in to their workstation. Logged the incident in ServiceNow, fixed the issue in Active Directory, then closed the ticket.

Steps to Set Up ServiceNow

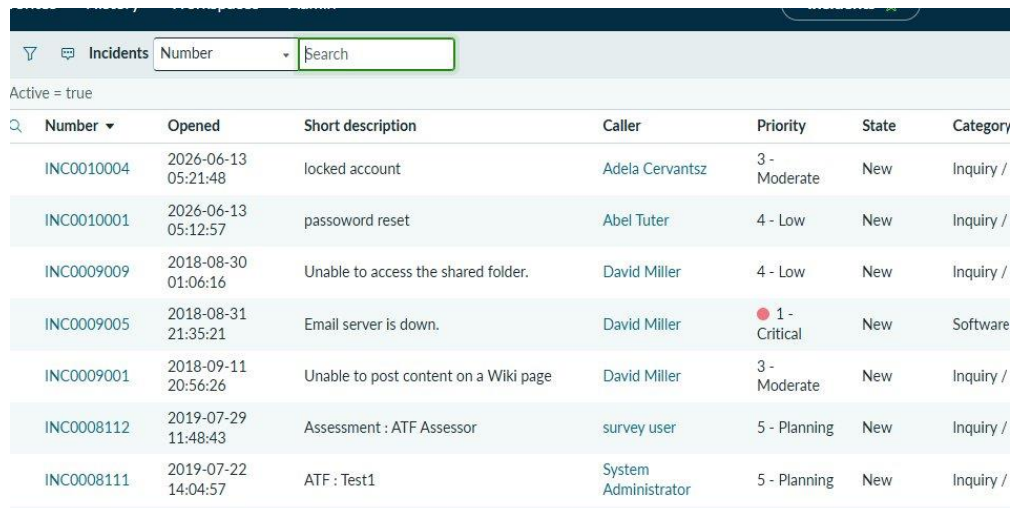
- Signed up for a free ServiceNow Developer Instance at developer.servicenow.com
- Logged in and navigated to the Incidents module under Service Desk in the left menu
- Familiarised with the incident queue showing all open tickets

Steps to Log the Incident

- Clicked New to create a new incident
- Set Caller to Emily Carter, Category to Inquiry/Help, Channel to Phone
- Entered Short description: workstation Locked out
- Entered Description: She said she cannot log in to her workstation. It needs resetting
- Set Urgency to 3 Low and assigned to Help Desk group then saved the ticket

Steps to Resolve the Issue

- Went to ADUC on the Domain Controller and searched for Emily Carter's account
- Right-clicked her account and saw it was disabled
- Clicked Enable Account to restore her login access
- Returned to ServiceNow and opened the ticket
- Changed State to Resolved and filled in the Resolution Information tab
- Clicked Resolve to close the ticket



Number	Opened	Short description	Caller	Priority	State	Category
INC0010004	2026-06-13 05:21:48	locked account	Adela Cervantsz	3 - Moderate	New	Inquiry /
INC0010001	2026-06-13 05:12:57	password reset	Abel Tuter	4 - Low	New	Inquiry /
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry /
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry /
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry /
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry /

Fig 10 - ServiceNow incident queue showing INC0010005

Number

INC0010005

Channel

Phone

* Caller

Emily Carter

Q

State

In Progress

Category

Inquiry / Help

The extent to which resolution of an incident can bear delay

Subcategory

-- None --

Urgency

3 - Low

Service

Priority

4 - Low

Service offering

Assignment group

Help Desk

Configuration item

CALLXPR1

Q

Assigned to

System Administrator

Q

* Short description

workstation Locked out

Description

She said she can't log in to her workstation. It needs resetting.

Related Search Results

Related Records

Resolution Information

Watch list

Work notes list

Work notes

Work notes

Fig 11 - ServiceNow ticket form showing the incident details

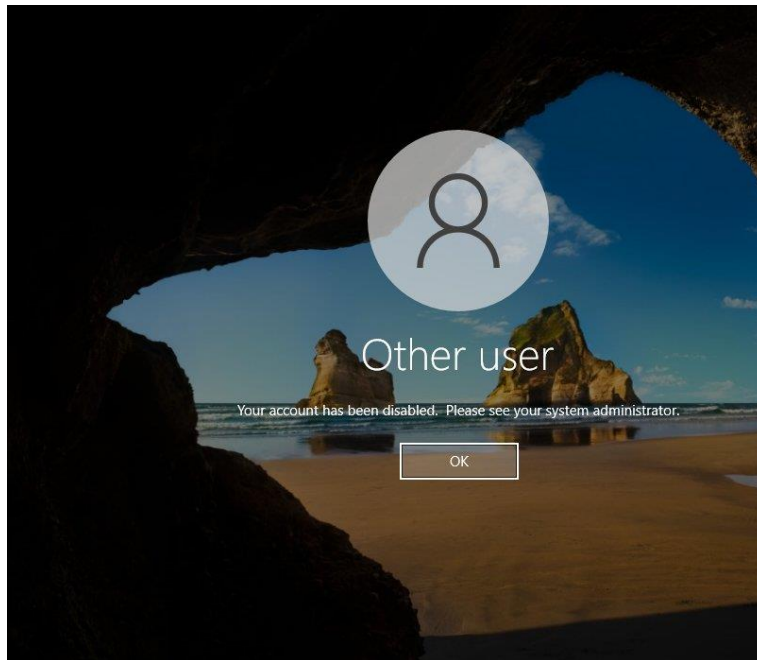


Fig 12 - The error Emily saw when trying to log in

State = Resolved											
Q	Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
	INC0010005	2026-06-13 05:36:14	workstation Locked out	Emily Carter	4 - Low	Resolved	Inquiry / Help	Help Desk	(empty)	2026-06-13 06:11:33	admin

Fig 13 - INC0010005 marked as Resolved in ServiceNow

Why This Matters

This covers the full ITIL incident lifecycle from logging through to closure. ServiceNow is the most widely used ITSM platform in enterprise environments and being able to work a ticket end to end is a core 1st line support skill.

08

Troubleshoot Trust Relationship Between Workstation and Domain Failed

Active Directory | Troubleshooting

User Daniel Hughes received the error 'The trust relationship between this workstation and the primary domain failed' when trying to log in. Diagnosed and resolved the issue.

Cause

Each domain-joined PC has a machine account in AD with a password that rotates every 30 days. If the PC is offline too long or restored from a snapshot, that password falls out of sync with the DC and breaks the trust.

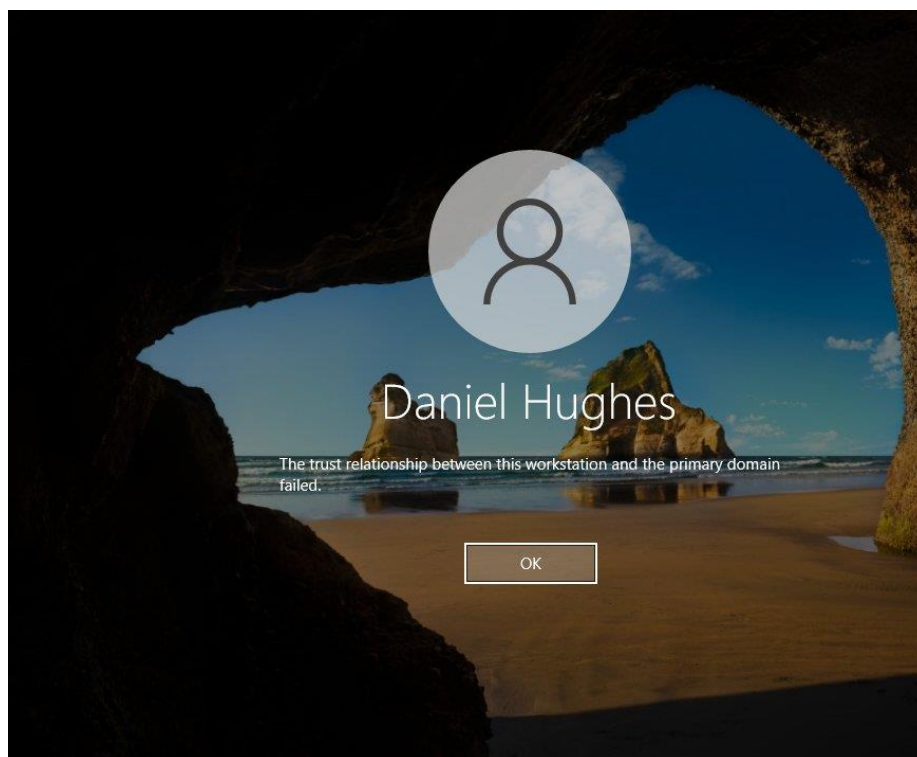


Fig 14 - Trust relationship error on the Windows 10 login screen

Steps to Fix

- At the Windows 10 login screen, clicked Other user and logged in with the local admin account instead of the domain account
- Opened PowerShell as Administrator by right-clicking the Start button and selecting Windows PowerShell (Admin)

- Typed the following command and pressed Enter: `Test-ComputerSecureChannel -Repair -Credential (Get-Credential)`
- A credentials box appeared, entered the Domain Admin username and password then clicked OK
- PowerShell returned True confirming the secure channel was reset successfully
- Alternative method if PowerShell did not work: opened ADUC on the DC, right-clicked the computer object for the affected machine and selected Reset Account
- Restarted the workstation and confirmed Daniel Hughes could log in with domain credentials

Why This Matters

This is a common error in environments with VMs or machines that go offline for long periods. Knowing how to fix it via PowerShell or ADUC is useful for any 1st or 2nd line support role.

09

GPO: Enforce Screensaver Timeout on HR Users

Group Policy | Security

Created a GPO called Screensaver policy, configured a screensaver timeout setting, and linked it to the HR OU so it applies to all HR users automatically.

Steps to Create and Configure the GPO

- Opened Group Policy Management from Server Manager under Tools
- Expanded Forest, Domains, mylab.local then right-clicked the HR OU
- Selected Create a GPO in this domain and link it here
- Named the GPO Screensaver policy and clicked OK
- Right-clicked the new GPO and selected Edit to open Group Policy Management Editor
- Navigated to User Configuration, Policies, Administrative Templates, Control Panel, Personalisation
- Double-clicked Screen saver timeout and selected Enabled
- Set the timeout value in seconds then clicked Apply and OK
- Closed the editor, the policy was now active and linked to the HR OU

Policy Details

- **GPO name:** Screensaver policy
- **Setting:** Screen saver timeout
- **Linked to:** HR OU (mylab.local/HR)

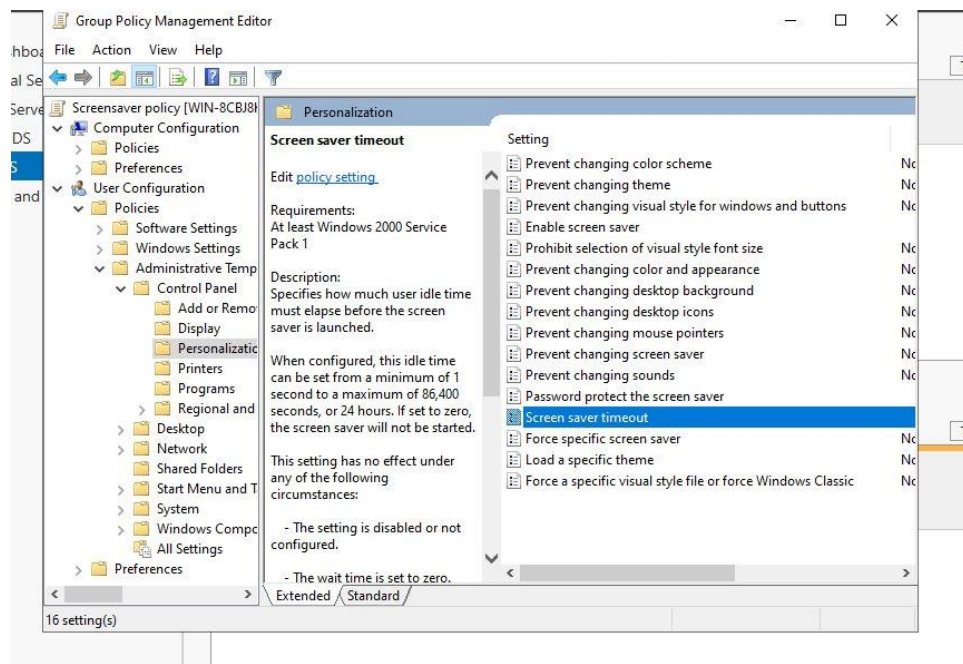


Fig 15 - Screen saver timeout setting in Group Policy Management Editor



Fig 16 - Screensaver policy linked to the HR OU

Why This Matters

Linking the GPO to the HR OU means only HR users are affected. A screensaver lockout is a standard security control used to prevent unauthorised access to idle workstations.

Created a GPO called Disable USB driver, enabled the setting to deny all removable storage access, and linked it to the Workstation OU to apply across all company PCs.

Steps

- Opened Group Policy Management from Server Manager under Tools
- Right-clicked the Workstation OU and selected Create a GPO in this domain and link it here
- Named the GPO Disable USB driver and clicked OK
- Right-clicked the GPO and selected Edit
- Navigated to Computer Configuration, Policies, Administrative Templates, System, Removable Storage Access
- Double-clicked All Removable Storage classes: Deny all access and selected Enabled
- Clicked Apply then OK and closed the editor
- Confirmed the GPO appeared as linked and enabled in the Workstation OU

Policy Details

- **GPO name:** Disable USB driver
- **Setting:** All Removable Storage classes: Deny all access
- **Status:** Enabled
- **Linked to:** Workstation OU

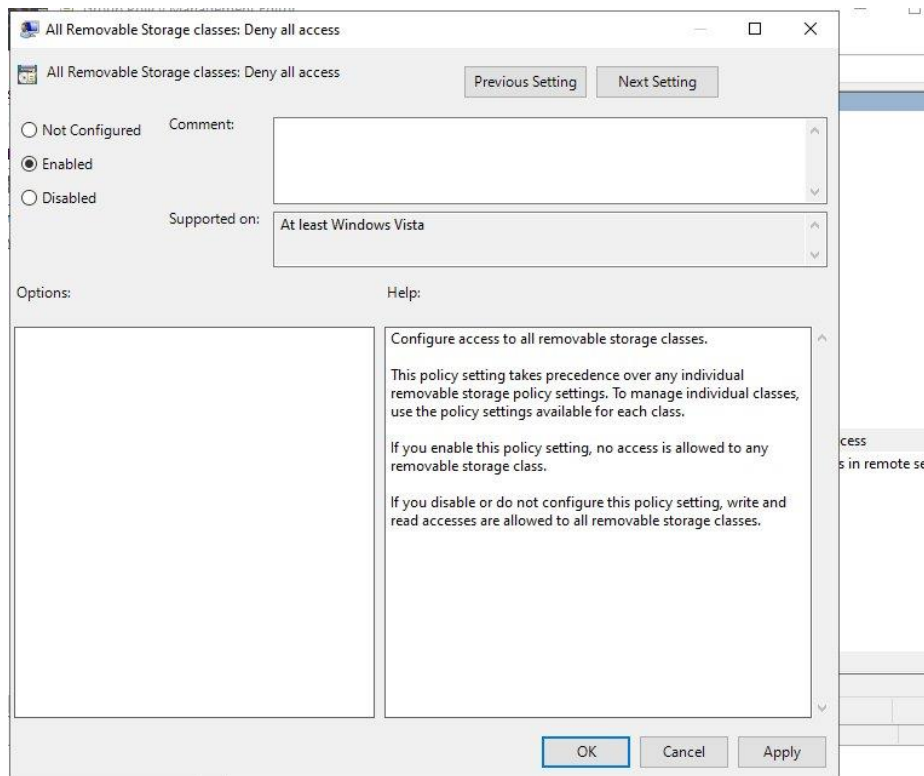


Fig 17 - Removable storage access denied setting enabled in GPO

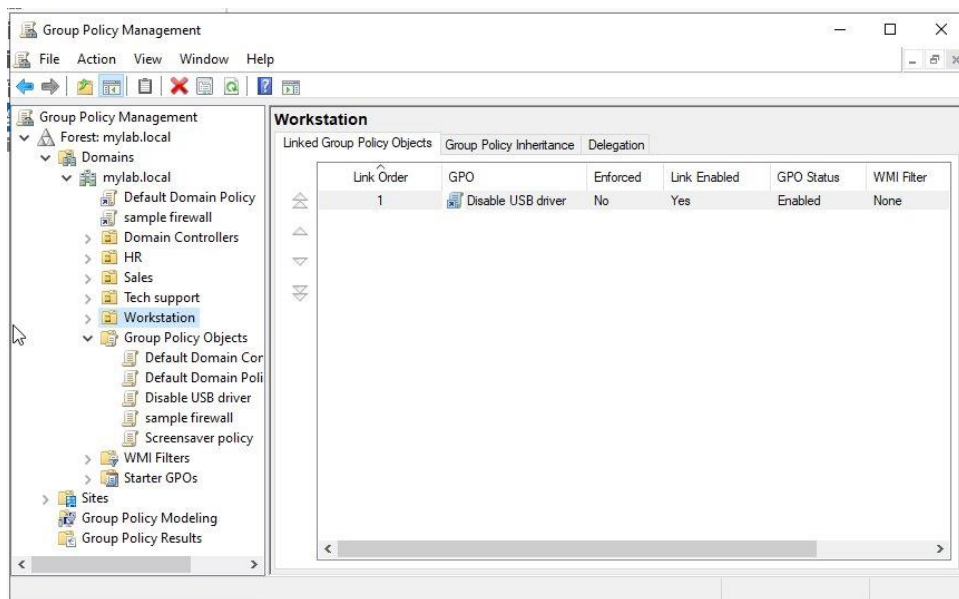


Fig 18 - Disable USB driver GPO linked to the Workstation OU

Why This Matters

Blocking removable storage prevents users copying sensitive data to USB drives or introducing malware. Applying it at OU level means it enforces on every PC in that OU without any per-machine configuration.

11

Create a Shared Folder with NTFS Permissions and Map as Network Drive

File Server | NTFS | Network Drive Mapping

Created an HR folder on the Domain Controller, restricted access using NTFS permissions tied to the HR Users AD group, then mapped it as a network drive on the Windows 10 client.

Steps to Create and Share the Folder

- On the Domain Controller, opened File Explorer and went to the C: drive
- Right-clicked and selected New, Folder and named it HR
- Right-clicked the HR folder and selected Properties then clicked the Sharing tab
- Clicked Advanced Sharing, ticked Share this folder and set the share name
- Clicked Permissions and added the HR Users group with Read and Change permissions
- Clicked OK to apply the share settings

Steps to Set NTFS Permissions

- Clicked the Security tab in the HR folder Properties
- Clicked Edit then Add and typed HR Users then clicked Check Names and OK
- Selected the HR Users group and ticked the permissions needed then clicked Apply and OK

Steps to Map the Drive on Windows 10

- On the Windows 10 client, opened File Explorer and clicked This PC
- Clicked the Computer tab at the top and selected Map network drive
- Selected drive letter H: and entered the network path to the HR folder on the server
- Ticked Reconnect at sign-in so the drive maps automatically every time the user logs in
- Clicked Finish and the HR file drive appeared under Network Locations

Permissions Applied

- SYSTEM: Full control
- HR Users (MYLAB\HR Users): Read and write access
- Administrators: Full control
- Anyone not listed is denied access

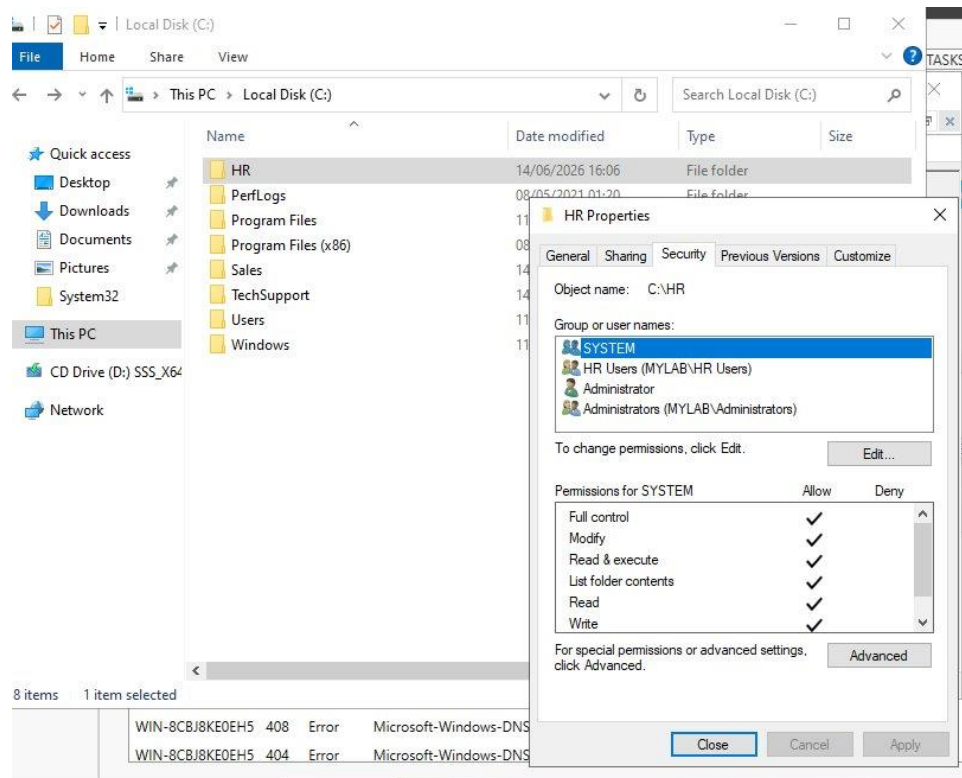


Fig 19 - HR folder NTFS permissions showing HR Users group access

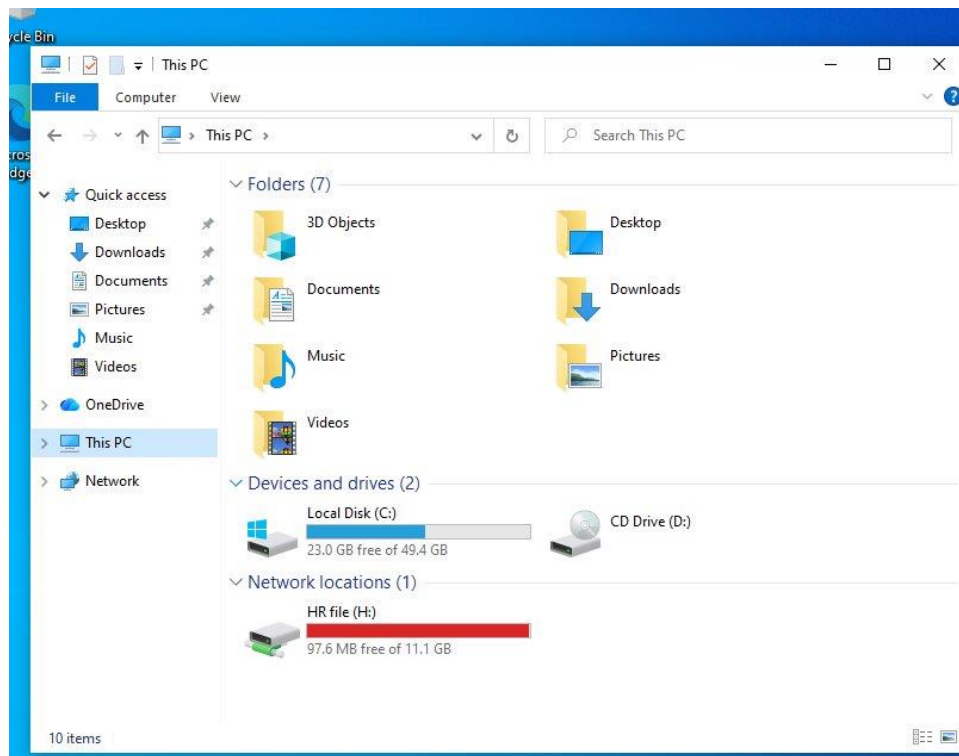


Fig 20 - Windows 10 showing the HR folder mapped as network drive H:

Why This Matters

Assigning permissions to the HR Users group keeps things manageable. New HR staff just need to be added to the group to get access. The mapped drive appears automatically at login, which is how departmental drives work in any company.

12

Configure Password Policy and Account Lockout in Default Domain Policy

Group Policy | Security | Password Policy

Edited the Default Domain Policy to enforce password complexity and account lockout rules that apply automatically to every user on the domain.

Steps to Configure Password Policy

- Opened Group Policy Management from Server Manager under Tools
- Expanded Forest, Domains, mylab.local and right-clicked Default Domain Policy then selected Edit
- Navigated to Computer Configuration, Policies, Windows Settings, Security Settings, Account Policies, Password Policy
- Double-clicked Enforce password history and set it to 10 passwords then clicked OK
- Double-clicked Maximum password age and set it to 90 days then clicked OK
- Double-clicked Minimum password age and set it to 1 day then clicked OK

- Double-clicked Minimum password length and set it to 7 characters then clicked OK
- Double-clicked Password must meet complexity requirements and selected Enabled then clicked OK

Steps to Configure Account Lockout Policy

- In the same editor, navigated to Account Policies then Account Lockout Policy
- Double-clicked Account lockout threshold and set it to 5 invalid attempts then clicked OK
- Windows automatically suggested values for the other settings, clicked OK to accept them
- Confirmed Account lockout duration was set to 30 minutes
- Confirmed Reset account lockout counter after was set to 15 minutes

Settings Applied

- **Password history:** 10 passwords remembered
- **Maximum age:** 90 days
- **Minimum length:** 7 characters
- **Complexity requirements:** Enabled
- **Lockout threshold:** 5 invalid attempts
- **Lockout duration:** 30 minutes
- **Reset counter after:** 15 minutes

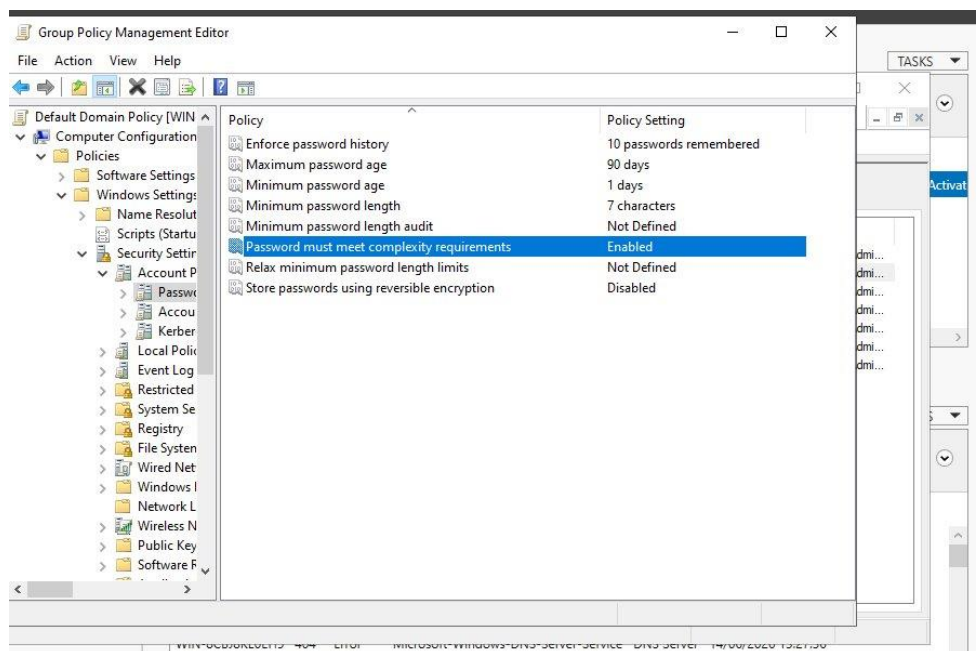


Fig 21 - Password policy settings in the Default Domain Policy

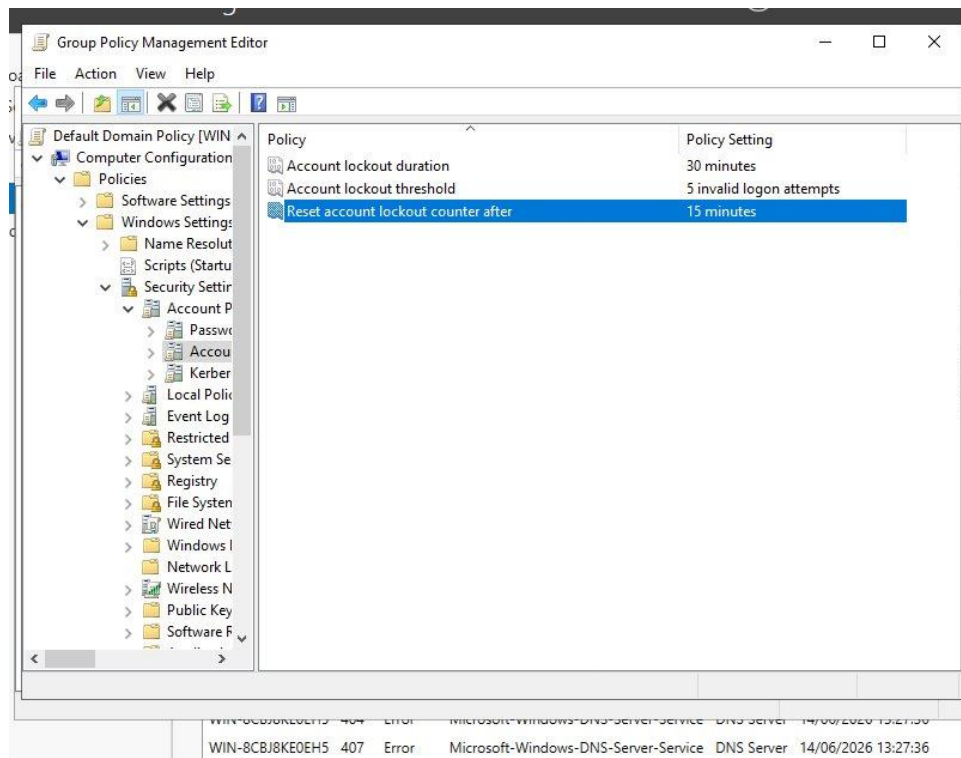


Fig 22 - Account lockout settings in the Default Domain Policy

Why This Matters

Complexity requirements stop users setting weak passwords. The lockout policy means an account locks after 5 wrong attempts, making brute force attacks impractical. Both policies apply to every user on the domain through the Default Domain Policy.

Microsoft 365 Administration

Microsoft 365 Admin Centre | Entra ID | Exchange Online | Intune

13

Create Users in Microsoft 365 Admin Centre

Microsoft 365 | User Management | Licensing

Created users in the Microsoft 365 Admin Centre using two methods — manually adding a single user and using a user template for the Sales team.

Steps to Create a User Manually

- Went to admin.microsoft.com and signed in as the global admin
- Clicked Users in the left menu then selected Active users
- Clicked Add a user at the top
- Entered First name: Ryan, Last name: Mitchell, Display name: Ryan Mitchell
- Set the username to Mryan and selected the domain hamzadahir2.onmicrosoft.com
- Left Automatically create a password ticked and ticked Require this user to change their password when they first sign in
- Clicked Next, assigned Microsoft 365 Business Standard licence, clicked Next then Finish adding

The screenshot shows the 'Set up the basics' form in the Microsoft 365 Admin Centre. On the left is a navigation pane with four steps: Basics (selected), Product licenses, Optional settings, and Finish. The main content area has the title 'Set up the basics' and a subtitle 'To get started, fill out some basic information about who you're adding as a user.' The form contains the following fields and options:

- First name:** Text input with 'Ryan' entered.
- Last name:** Text input with 'Mitchell' entered.
- Display name:** Text input with 'Ryan Mitchell' entered.
- Username:** Text input with 'Mryan' entered.
- Domains:** A dropdown menu showing '@ hamzadahir2.onmicrosoft.com'.
- Automatically create a password:** A checked checkbox.
- Require this user to change their password when they first sign in:** A checked checkbox.

Fig 23 - Creating user Ryan Mitchell in the Microsoft 365 Admin Centre

Steps to Create a User Template

- In the Admin Centre clicked Users then User templates
- Clicked Add a template and entered the name as Sales with a description
- Set the domain to hamzadahir2.onmicrosoft.com and configured password settings
- Assigned Microsoft 365 Business Standard licence to the template
- Reviewed all settings on the final screen and clicked Create template
- Template published and available for onboarding new sales staff quickly

Add user template

Description

Basics

Licenses

Optional settings

Finish

Review and finish creating your template

Review the settings for your template. You can use this template immediately after you finish creating it.

Template description

Name: Sales
Description: this template is for sales team
Publish status: Published
[Edit](#)

Domain

hamzadahir2.onmicrosoft.com
[Edit](#)

Password

Type: Auto-generated
Require users to change password on first login
[Edit](#)

Product licenses

Location: GB
Licenses: Microsoft 365 Business Standard
Apps: Avatars for Teams, Avatars for Teams (additional), Common Data Service, and 41 more
[Edit](#)

Roles

Fig 24 - Sales user template review page showing licence and password settings

Why This Matters

Templates save time when onboarding a whole team. Every user created from the template gets the same licence and settings automatically.

14

Create a Microsoft 365 Group and Add Members

Microsoft 365 | Groups | Collaboration

Created a Microsoft 365 group called HR, set an owner, added members and configured the group settings.

Steps

- In the Admin Centre clicked Teams and groups then Active teams and groups
- Clicked Add a group and selected Microsoft 365 as the group type then clicked Next
- Entered the group name as HR and added a description then clicked Next
- Assigned an owner for the group and clicked Next
- Added HR users as members by searching their names and clicking Add then Next
- Configured whether external senders could email the group on the Settings page
- Clicked Create group then Close once created
- The HR group appeared in the Active teams and groups list within a few minutes

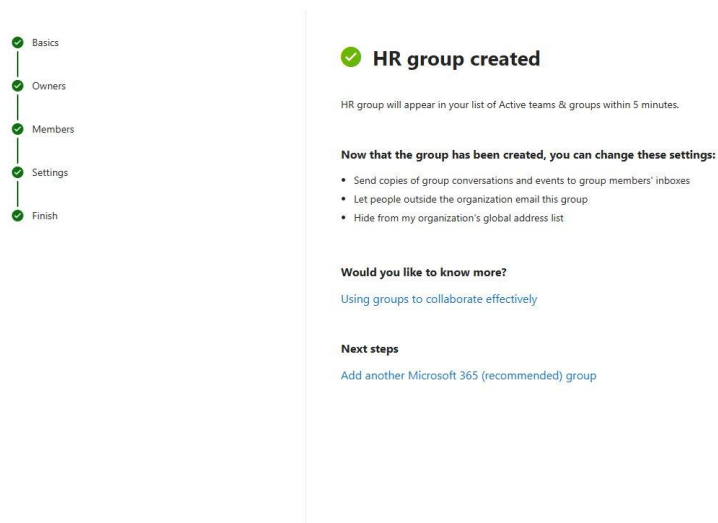


Fig 25 - HR group successfully created in Microsoft 365 Admin Centre

Why This Matters

A single Microsoft 365 group gives members shared access to a mailbox, Teams channel, SharePoint site and calendar all at once.

15

Verify Group Access Controls Across Mailbox, Teams and SharePoint

Microsoft 365 | Access Control | Permissions Testing

Tested the Sales group permissions by logging in as a group member and as a non-member to confirm access was controlled correctly.

Steps to Test as a Group Member

- Opened a private browser window and went to outlook.office.com
- Signed in as Laura Bennett (Blaura@hamzadahir2.onmicrosoft.com)
- Checked the inbox and received the Sales Team welcome email confirming membership
- The shared Sales mailbox appeared in the left panel under her personal mailbox
- Navigated to the Sales SharePoint site and confirmed she had full access

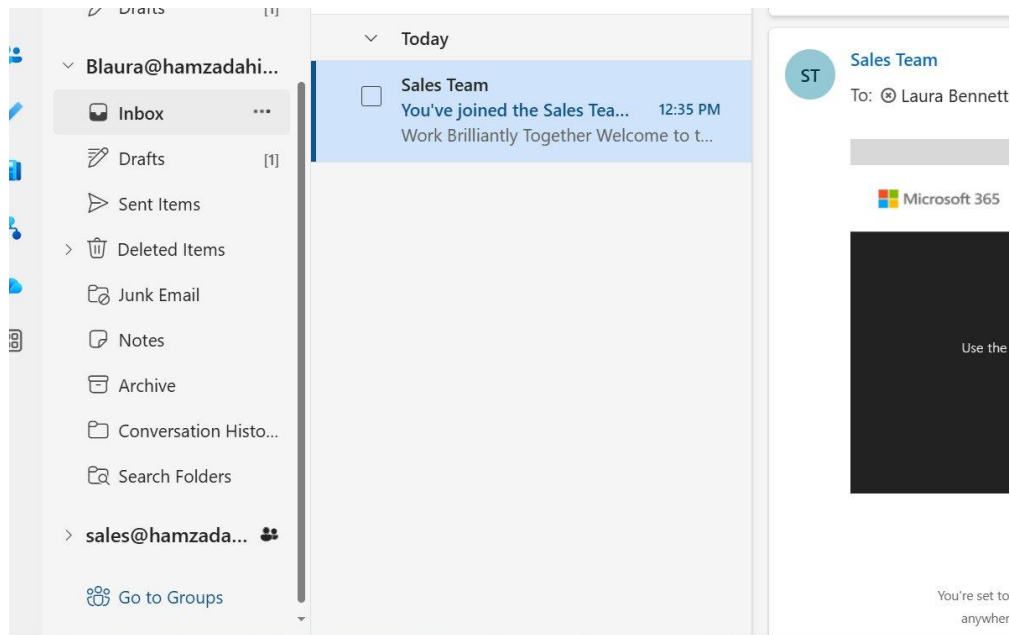


Fig 26 - Laura Bennett's inbox showing the Sales Team welcome email and shared mailbox

Steps to Test as a Non-Member

- Opened another private browser window and signed in as a user not in the Sales group
- Attempted to navigate directly to the Sales SharePoint site URL
- Received the You need access page confirming the user was blocked
- Confirmed the user could request access but could not view any content

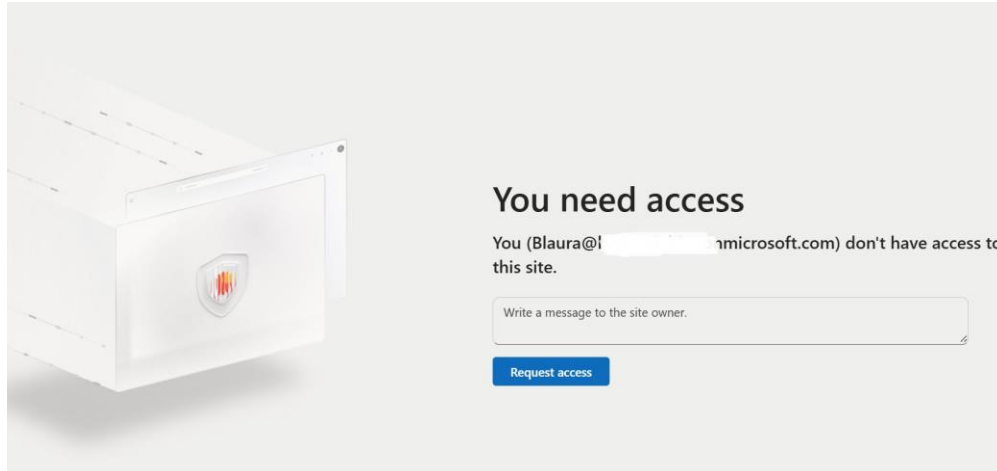


Fig 27 - Access denied page confirming non-members cannot access the Sales site

Why This Matters

Testing as both a member and non-member confirms permissions are working and departments are kept separate.

Created a company-wide distribution list and configured an automatic out of office reply on behalf of a user.

Steps to Create the Distribution List

- In the Admin Centre clicked Teams and groups then Active teams and groups
- Clicked the Distribution list tab at the top then clicked Add a group
- Selected Distribution as the type, named it Allmembers and set the group email address
- Added all company users as members and clicked Create
- The Allmembers distribution list appeared and was ready to use

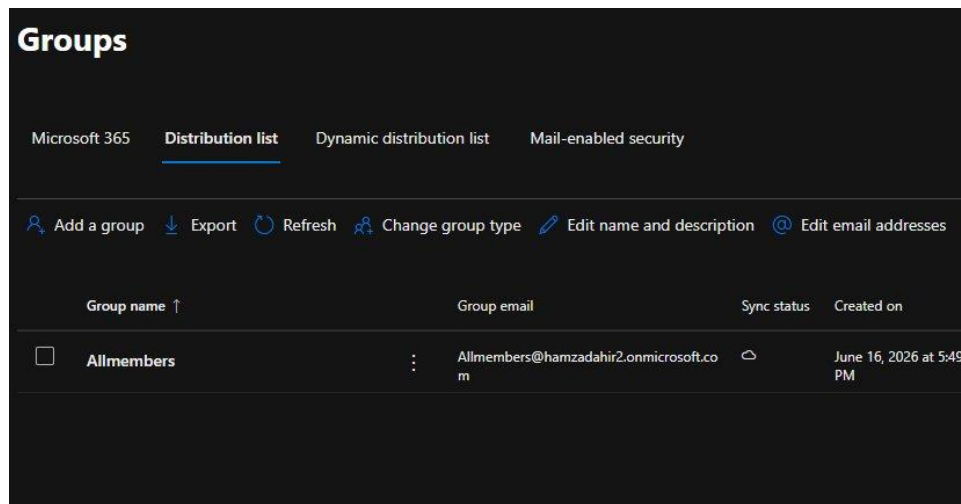


Fig 28 - Allmembers distribution list created in the Microsoft 365 Admin Centre

Steps to Set Up Automatic Replies

- In the Admin Centre clicked Users then Active users and selected the user
- Clicked the Mail tab and then clicked Manage automatic replies
- Toggled Automatic replies to On
- Ticked Send replies only during this time period and set the start date to 16/06/2026 and end date to 17/06/2026
- Typed the internal reply message and ticked Send automatic replies to senders outside this organization
- Selected Reply to all senders and typed the external reply message then clicked Save

←

Manage automatic replies

Automatic replies On

☒ Send replies only during this time period

Start date: 6/16/2026 - 1:00 AM

End date: 6/17/2026 - 1:00 AM

Reply to all senders inside the organizations from this mailbox

I am currently out of office, please contact
hr@yourdomain.onmicrosoft.com

*This field cannot be empty if automatic reply is on. 127926 characters left

☒ Send automatic replies to senders outside this organization

☐ Only reply to senders in this mailbox's contact list

☒ Reply to all senders

Reply to all senders outside the organizations from this mailbox

I am currently out of office, please contact
hr@yourdomain.onmicrosoft.com

*This field cannot be empty if automatic reply is on. 127926 characters left

Save

Fig 29 - Automatic replies configured with internal and external messages

Why This Matters

Setting automatic replies on behalf of a user is a common helpdesk task when someone is off sick and forgot to set it themselves.

17

Configure Email Forwarding Between Users

Exchange Online | Mailbox Management

Set up email forwarding on a user's mailbox so all incoming emails are automatically redirected to Ryan Mitchell.

Steps

- In the Admin Centre clicked Users then Active users and selected the user to forward from

- Clicked the Mail tab and scrolled down to Email forwarding then clicked Manage email forwarding
- Toggled Forward all emails sent to this mailbox to On
- Selected Forward to an internal email address and searched for Ryan Mitchell
- Left Deliver message to both mailboxes unticked so emails go to Ryan only
- Clicked Save to apply the forwarding rule

Manage email forwarding

You can use this setting to forward email messages sent to the user to another recipient. [Learn more](#)

Forward all emails sent to this mailbox ☒

Forward all emails sent to this mailbox
[Learn more about controlling automatic external email forwarding](#)

Automatic email forwarding to external recipients makes your organization vulnerable to takeover attacks. If you choose to do it anyway, you can create an anti-spam outbound policy that allows it only for users who need it.
[Manage outbound anti-spam policies](#)

Please note that if you set up email auto-forwarding to an external address outside your organisation, the mailbox owner will be able to see and change the forwarding address. However, if the forwarding is to an internal email address, the mailbox owner will not have access to modify it.

Forwarding address *

☒ Forward to an internal email address

☐ Forward to an external email address

☐ Deliver message to both forwarding address and mailbox

Fig 30 - Email forwarding configured to redirect all emails to Ryan Mitchell

Why This Matters

Email forwarding is used when someone leaves or changes role. Forwarding internally is more secure than forwarding to an external address.

Ryan Mitchell could not access a SharePoint site he needed. Logged the ticket, diagnosed the cause as missing group membership, fixed it and closed the ticket.

Steps to Log the Incident in ServiceNow

- Opened ServiceNow and navigated to Service Desk then Incidents
- Clicked New and set Caller to Emily Carter, Category to Inquiry/Help, Service to IT Services
- Entered Short description: Staff Member Unable to Access SharePoint Site
- Filled in the full description of the issue and set Priority to 4 Low then saved

Steps to Diagnose

- Opened the Microsoft 365 Admin Centre and went to Teams and groups
- Opened the Sales group and clicked the Members tab
- Confirmed Ryan Mitchell was not listed as a member
- Tested by signing in as Ryan and navigating to the Sales SharePoint site
- Received the You need access page confirming the diagnosis

⊗ The following mandatory fields are not filled in: Resolution code, Resolution notes

Number	INC0010007	Channel	-- None --
* Caller	Emily Carter	State	Resolved
Category	Inquiry / Help	Impact	3 - Low
Subcategory	-- None --	Urgency	3 - Low
Service	IT Services	Priority	4 - Low
Service offering		Assignment group	
Configuration item	.NET Framework	Assigned to	
* Short description	Staff Member Unable to Access SharePoint Site		
Description	A staff member, Ryan Mitchell, is unable to access a SharePoint site that he should have permission to use. User affected: Ryan Mitchell - Issue: Cannot access shared SharePoint site - Impact: Unable to view/work with shared files needed for daily tasks		

Fig 31 - ServiceNow incident INC0010007 logged with full details

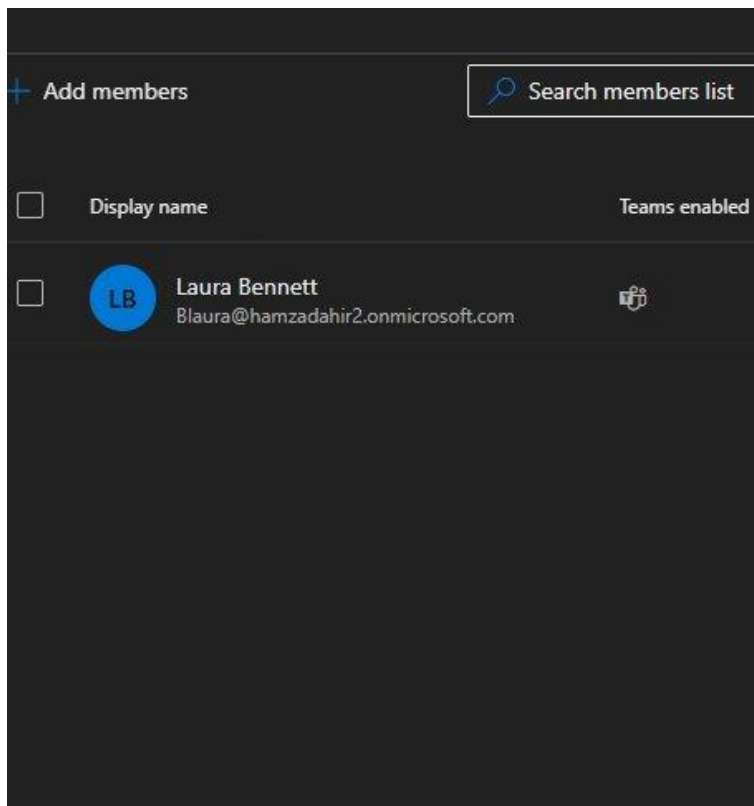


Fig 32 - Sales group showing only Laura Bennett, confirming Ryan was missing

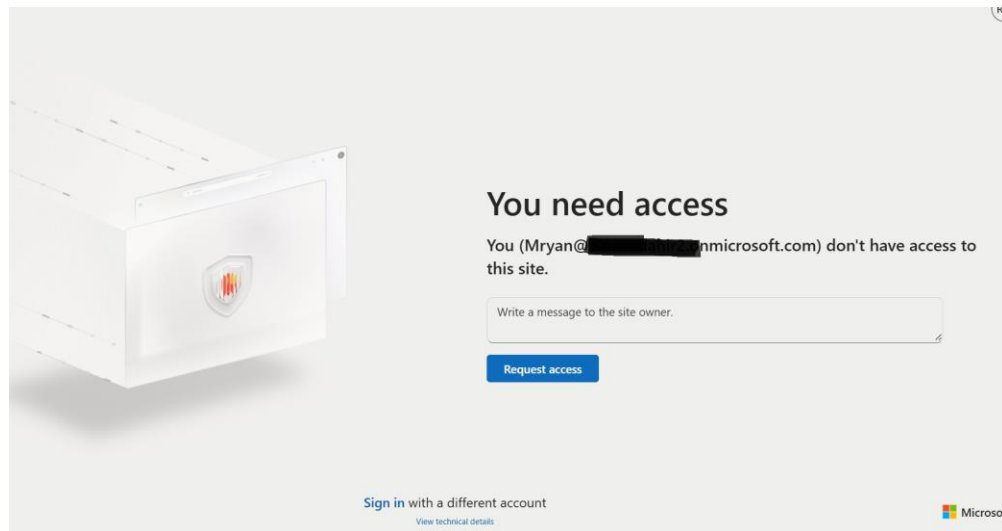


Fig 33 - Access denied page confirming Ryan Mitchell did not have site access

Steps to Fix and Close

- In the Sales group Members tab clicked Add members and searched for Ryan Mitchell
- Selected Ryan and clicked Add to add him to the group
- Waited a few minutes then confirmed Ryan could access the SharePoint site
- Returned to ServiceNow, opened the ticket and clicked Resolution Information tab
- Set Resolution code to Resolved by change and entered resolution notes explaining the fix
- Clicked Resolve to close the ticket

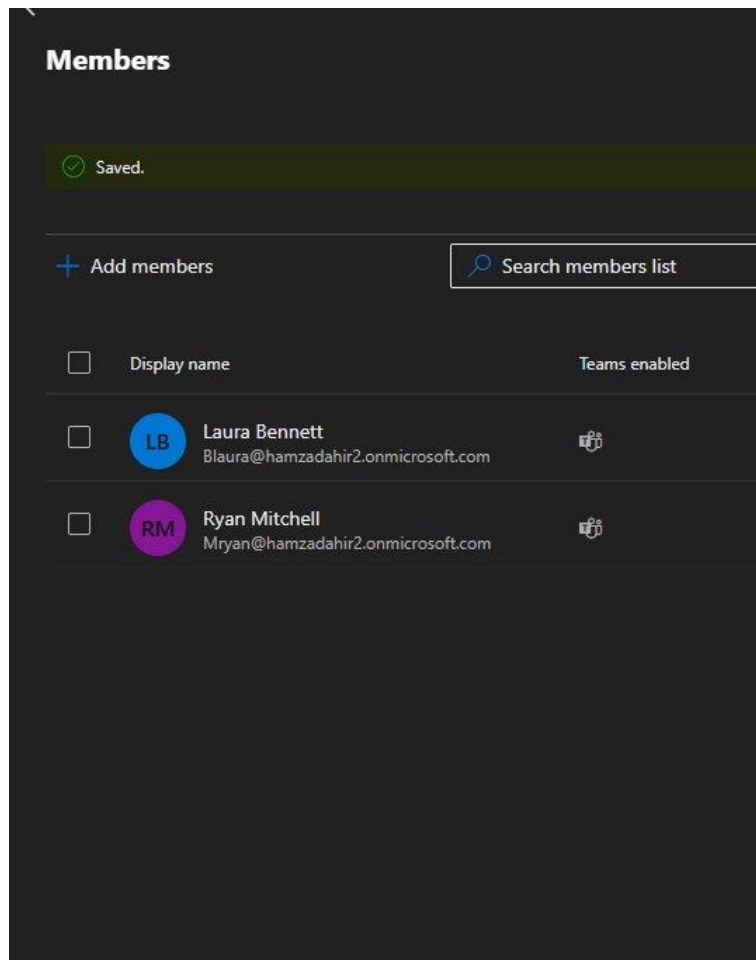


Fig 34 - Sales group members updated to include Ryan Mitchell

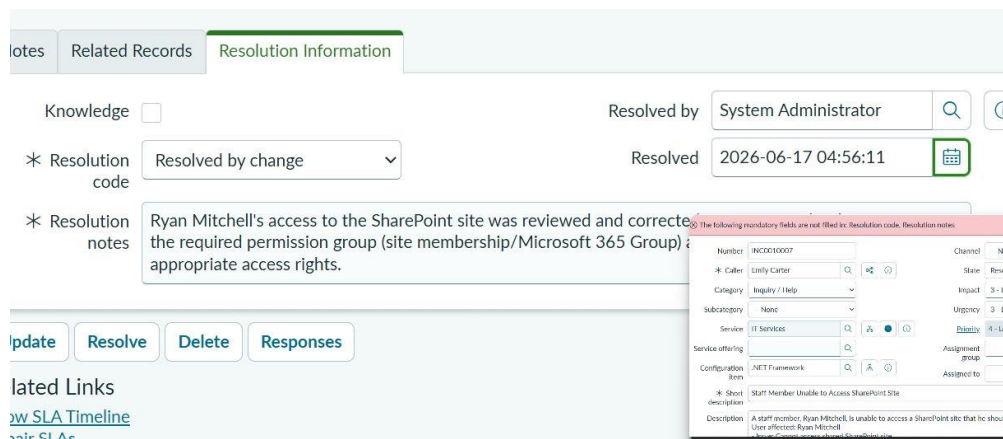


Fig 35 - Ticket resolved with resolution notes recorded

Why This Matters

Diagnosing by testing as the affected user, fixing the root cause and documenting it properly in the ticket are the steps expected of any support engineer.

A user deleted files from OneDrive and emptied their recycle bin. Recovered the files using the SharePoint second-stage recycle bin.

Steps

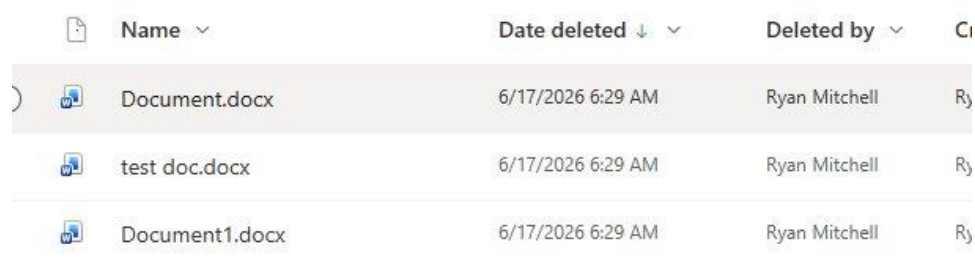
- Opened the user's OneDrive in a browser and clicked Recycle bin in the left panel
- The files were not there as the user had already emptied the recycle bin
- Scrolled to the bottom of the recycle bin page and clicked Check the second-stage recycle bin
- The second-stage recycle bin opened showing all recently deleted files
- Ticked the files to recover: Document.docx, test doc.docx and Document1.docx
- Clicked Restore at the top and confirmed the action
- Navigated back to OneDrive and confirmed the files were restored to their original location



File not found? Check the [Second-stage recycle bin](#)

Fig 36 - Link to the second-stage recycle bin when a file is missing from the normal one

Second stage recycle bin



	Name ▾	Date deleted ↓ ▾	Deleted by ▾	Can be restored
☒	Document.docx	6/17/2026 6:29 AM	Ryan Mitchell	Yes
☒	test doc.docx	6/17/2026 6:29 AM	Ryan Mitchell	Yes
☒	Document1.docx	6/17/2026 6:29 AM	Ryan Mitchell	Yes

Fig 37 - Second-stage recycle bin showing deleted files ready to restore

OneDrive and SharePoint have two stages of recycle bin so files can often be recovered even after a user thinks they are permanently deleted.

20

Join a Windows 10 Device to Entra ID and Enrol in Intune

Microsoft Entra ID | Intune | Device Management

Joined a Windows 10 machine to the Microsoft 365 organisation through Entra ID and confirmed it was enrolled and managed by Intune.

Steps to Join the Device to Entra ID

- On the Windows 10 machine opened Settings and clicked Accounts
- Clicked Access work or school then clicked Connect
- On the Set up a work or school account screen, clicked Join this device to Microsoft Entra ID
- Entered the Microsoft 365 credentials for Cemily@hamzadahir2.onmicrosoft.com
- Reviewed the Make sure this is your organization screen showing the tenant and user details
- Clicked Join to complete the Entra ID join

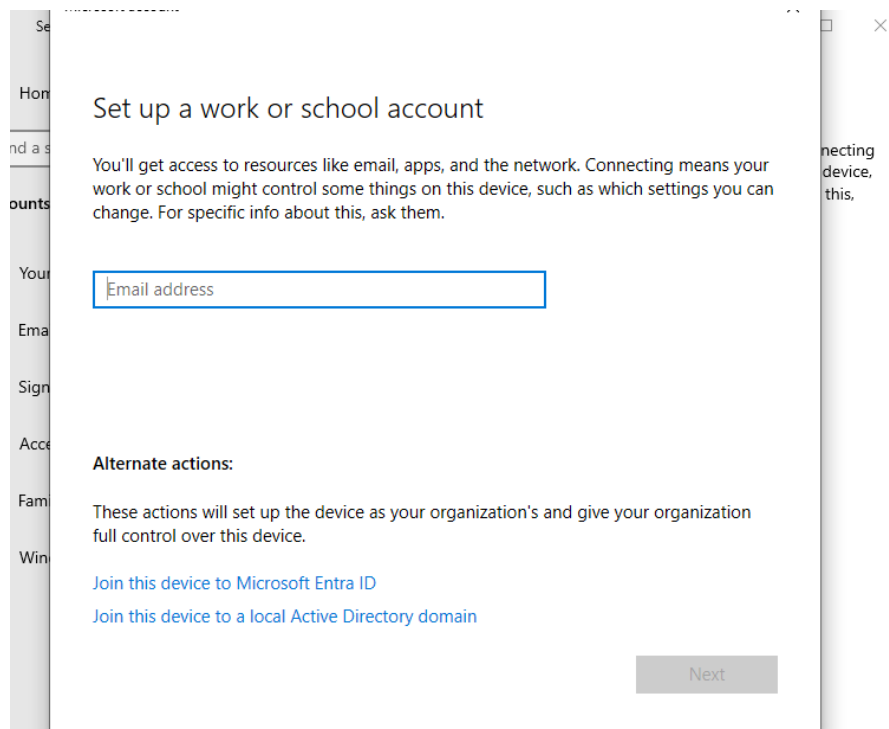


Fig 38 - Set up a work or school account screen showing the Entra ID join option

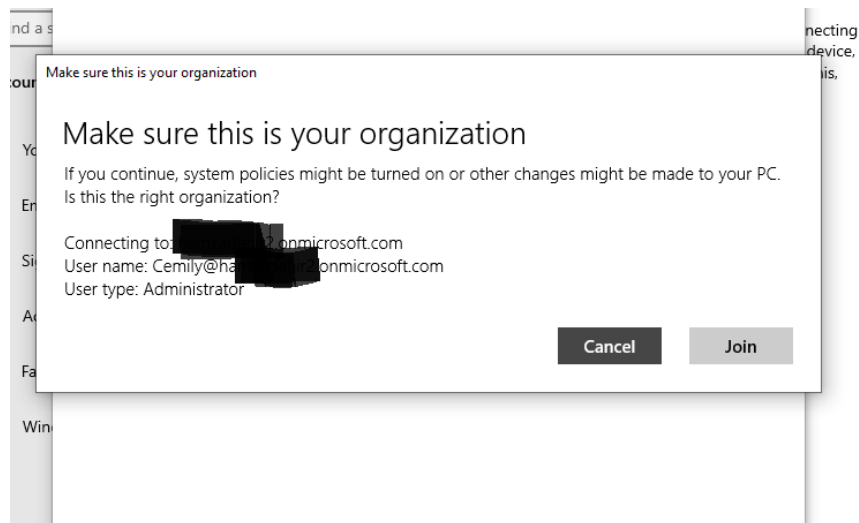


Fig 39 - Organisation confirmation screen before clicking Join

Verified in Microsoft Intune Admin Centre

- Opened endpoint.microsoft.com and navigated to Devices
- DESKTOP-9IJ5MMA appeared in the device list confirming successful enrolment
- Managed by: Intune
- Ownership: Corporate
- Compliance: Compliant

Refresh Export Columns Bulk device actions				
Search OS: Windows, Windows Mobile, Windows Holographic				
Device name	Managed by	Ownership	Compliance	OS
DESKTOP-9IJ5MMA	Intune	Corporate	Compliant	Windows

Fig 40 - Intune device list showing DESKTOP-9IJ5MMA as compliant and managed by Intune

Verified on the Device via Command Prompt

- Opened Command Prompt on the device and ran: `dsregcmd /status`
- Confirmed AzureAdJoined: YES
- Confirmed DomainJoined: NO (cloud only, not on-prem AD)

- Device Name: DESKTOP-9IJ5MMA

```
C:\Users\EmilyCarter>dsregcmd /status

-----
Device State
-----

        AzureAdJoined : YES
        EnterpriseJoined : NO
        DomainJoined : NO
        Device Name : DESKTOP-9IJ5MMA
-----
```

Fig 41 - dsregcmd /status output confirming the device is Azure AD joined

Why This Matters

Joining devices to Entra ID and managing them through Intune is how modern cloud-first organisations manage their fleet without needing an on-premises server. Intune can push policies, deploy apps and enforce compliance rules remotely. This is one of the most in-demand skills in IT support and platform engineering roles right now.

More tasks will be added as the lab progresses | Hamsa Dahir | June 2026

21 | Register a Device with Windows Autopilot

Microsoft Intune | Windows Autopilot | Device Enrolment

Registered a Windows 10 VM into Windows Autopilot by extracting the hardware ID using PowerShell, importing it into Intune, creating a deployment profile and confirming the device was ready to deploy.

Step 1 - Extract the Hardware ID Using PowerShell

- Opened PowerShell as Administrator on the Windows 10 VM
- Ran: `[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12` to set the security protocol
- Created a new folder at C:\HWID to store the output file
- Set execution policy to RemoteSigned to allow the script to run
- Installed the Get-WindowsAutopilotInfo script from PowerShell Gallery, confirmed Yes when prompted
- Ran: `Get-WindowsAutopilotInfo -OutputFile AutopilotHWID.csv`
- The hardware ID was exported as a CSV file saved to C:\HWID\AutopilotHWID.csv

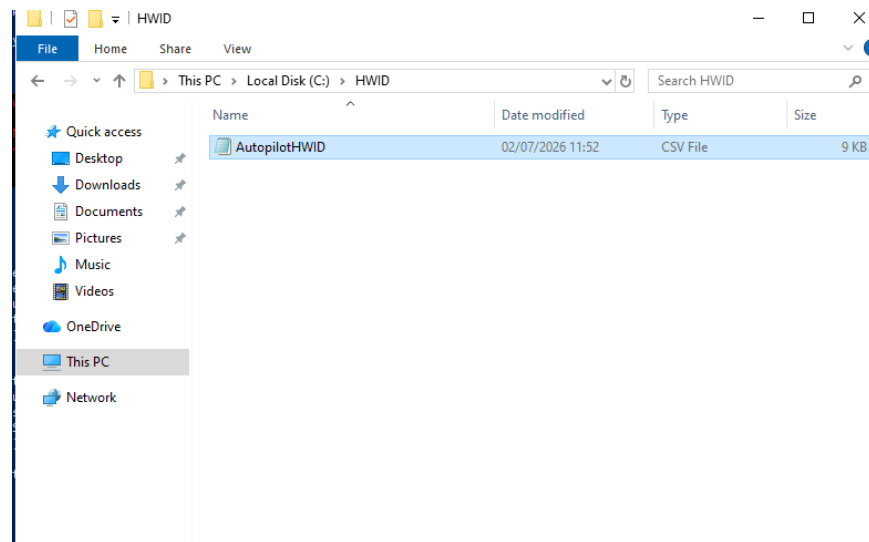


Fig 42 - AutopilotHWID.csv file saved to C:\HWID folder

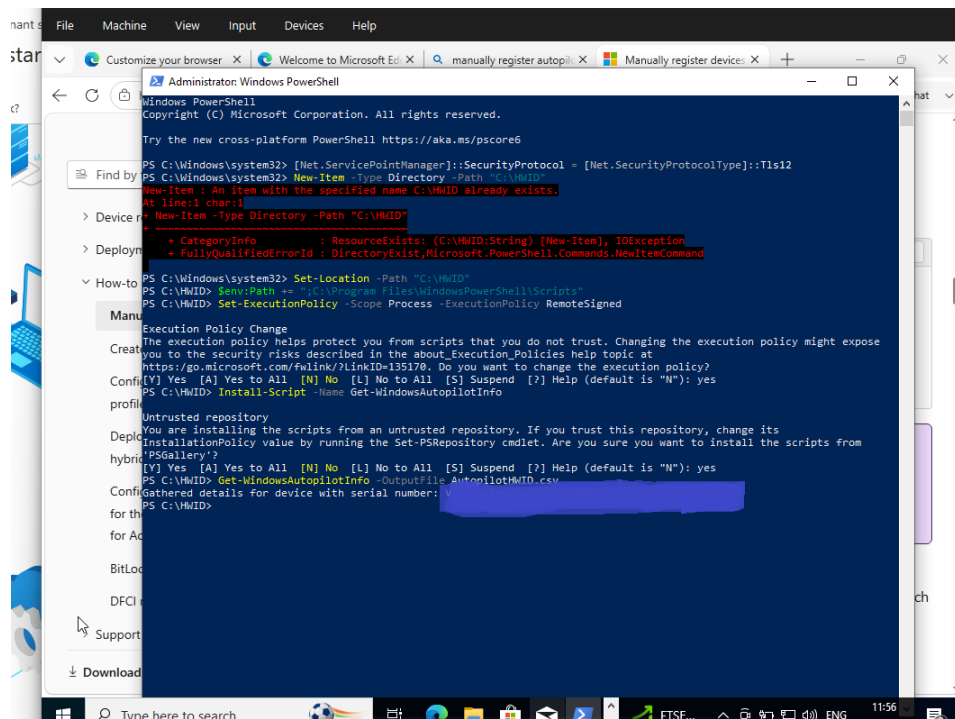


Fig 43 - PowerShell commands used to extract the device hardware ID

Step 2 - Import the Device into Autopilot

- Opened Intune Admin Centre at endpoint.microsoft.com
- Navigated to Devices, Windows, Windows Enrollment then Devices under Autopilot
- Clicked Import and uploaded the AutopilotHWID.csv file
- Initially showed No data found while the sync was processing
- Clicked Sync and waited for the device to appear
- Device registered successfully showing Serial number, Manufacturer: innotek GmbH and Model: VirtualBox

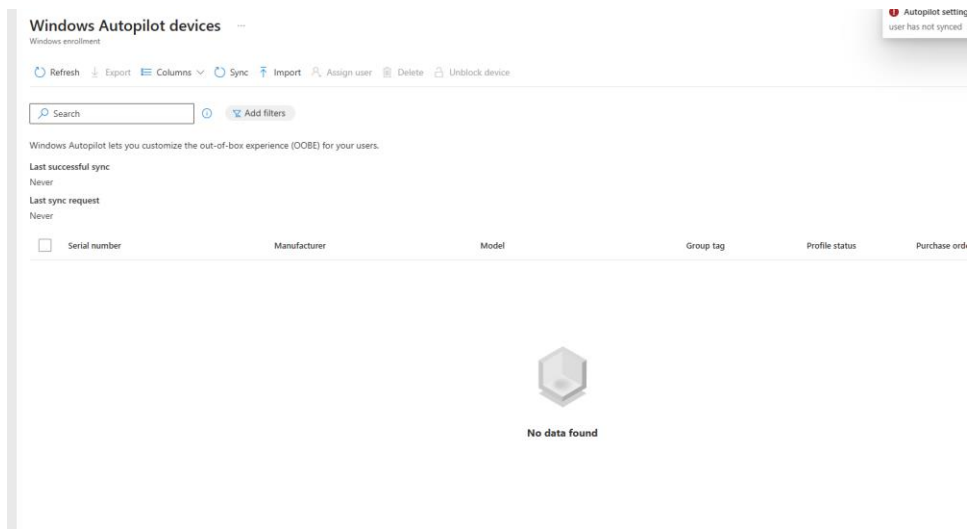


Fig 44 - Autopilot devices page before sync showing no data found

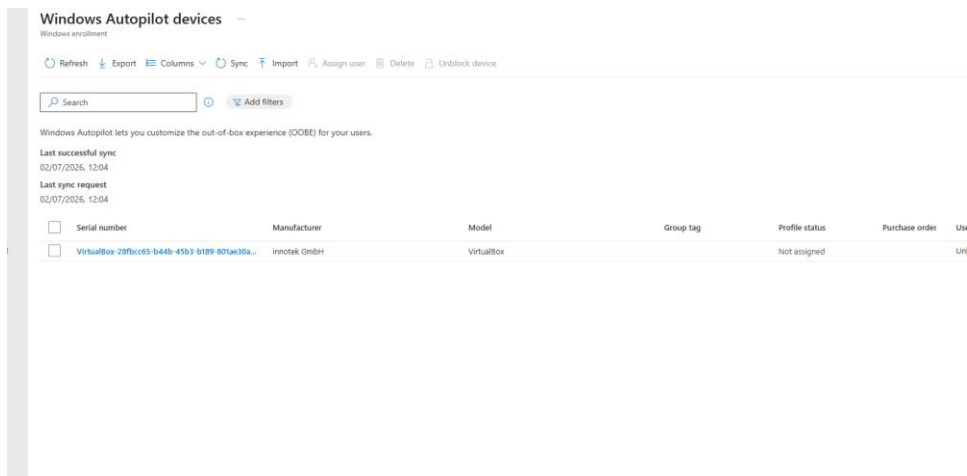


Fig 45 - Device successfully registered in Windows Autopilot after sync

Step 3 - Create a Windows Autopilot Deployment Profile

- Navigated to Devices, Windows Enrollment, Deployment Profiles and clicked Create Profile
- Set Deployment mode to User-Driven
- Set Join to Microsoft Entra ID as: Microsoft Entra joined
- Set Microsoft Software License Terms to Hide
- Set Privacy settings to Hide
- Set Hide change account options to Hide
- Set User account type to Standard (not Administrator)
- Enabled Apply device name template and set naming convention to MYLAB-%RAND:3% to auto-generate unique device names
- Clicked Next and completed the profile creation

Create profile ...

Windows PC

Configure the out-of-box experience for your Autopilot devices

Deployment mode *

Join to Microsoft Entra ID as *

Microsoft Software License Terms

[Important information about hiding license terms](#)

Privacy settings

[The default value for diagnostic data collection has changed for devices running Windows 10, version 1903 and later, or Windows 11.](#)

Hide change account options

User account type

Allow pre-provisioned deployment

Language (Region)

Automatically configure keyboard

Apply device name template

Create a unique name for your devices. Names must be 15 characters or less, and can contain letters (a-z, A-Z), numbers (0-9), and hyphens. Names must not contain only numbers. Names cannot include a blank space. Use the %SERIAL% macro to add a hardware-specific serial number. Alternatively, use the %RANDx% macro to add a random string of numbers, where x equals the number of digits to add.

Enter a name *

Fig 46 - Autopilot deployment profile configured with User-Driven mode and device name template

Step 4 - Confirmed on the Device

- Logged into the VM as Emily Carter and confirmed the device was connected to the organisation
- Microsoft 365 apps were available and the device was fully set up under the work account

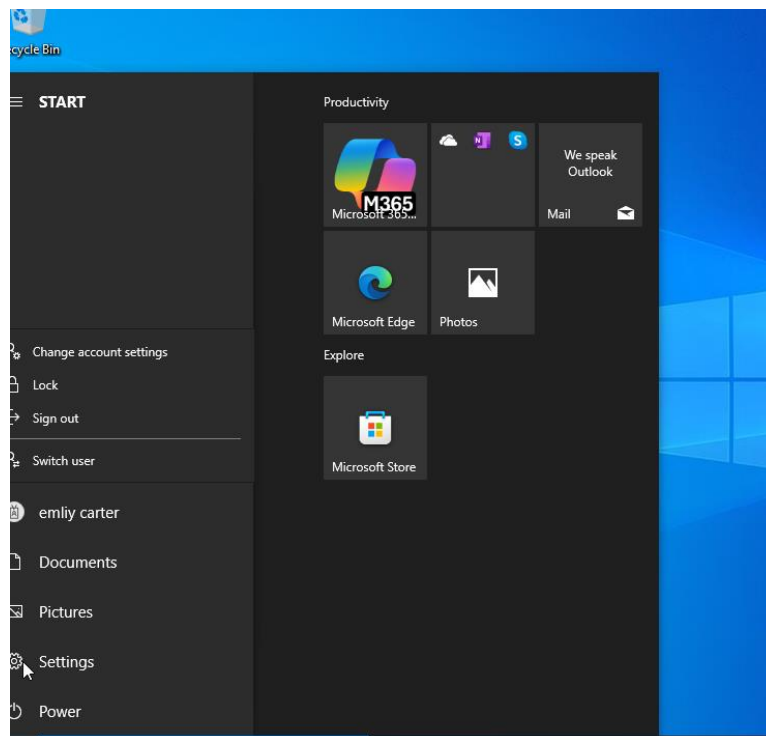


Fig 47 - Windows 10 Start menu showing Emily Carter logged in with Microsoft 365 apps available

Why This Matters

Windows Autopilot is how modern IT teams deploy new devices without manually imaging or configuring them. A device can ship directly to an employee, they power it on, sign in with their work account and Autopilot automatically sets everything up. The deployment profile controls what the user sees and what gets configured. This is a highly valued skill in platform engineering and IT support roles.

Task 22 | Create a Device Compliance Policy in Intune

Microsoft Intune | Compliance | Device Management

Created a compliance policy in Microsoft Intune that defines the minimum security requirements a Windows 10 device must meet to be considered compliant. The policy was assigned to all devices in the organisation.

Steps

- In Intune Admin Centre went to Devices then Compliance and clicked Create Policy
- Selected Windows 10 and later as the platform and clicked Create
- Named the policy and moved to Compliance settings
- Under System Security, Password: set Require a password to unlock mobile devices to Require
- Set Simple passwords to Block
- Set Password type to Device default
- Set Minimum password length to 7 characters
- Set Password expiration to 41 days
- Under Device Security: set Firewall to Require and Antivirus to Require
- Clicked Next through Actions for noncompliance and Assignments
- Assigned the policy to All devices and clicked Create

JUN 15, 10:41 AM

Basics **Compliance settings** Actions for noncompliance 4 Assignments 5 Review + create

Custom Compliance

Custom compliance ⓘ Require Not configured

Select your discovery script [Click to select](#)

Upload and validate the JSON file with your custom compliance settings

Device Health

Device Properties

Configuration Manager Compliance

System Security

Password

Require a password to unlock mobile devices ⓘ Require Not configured

Simple passwords ⓘ Block Not configured

Password type ⓘ Device default ▼

Minimum password length ⓘ 7 ✓

Maximum minutes of inactivity before password is required ⓘ Not configured ▼

Password expiration (days) ⓘ 41 ✓

Fig 48 - Compliance policy settings showing password and system security configuration

Encryption

Require encryption of data storage on device. ⓘ

Require Not configured

Device Security

Firewall ⓘ

Require Not configured

Trusted Platform Module (TPM) ⓘ

Require Not configured

Antivirus ⓘ

Require Not configured

Antispyware ⓘ

Require Not configured

Defender

Microsoft Defender Antimalware ⓘ

Require Not configured

Microsoft Defender Antimalware minimum version ⓘ

Not configured

Microsoft Defender Antimalware security intelligence up-to-date ⓘ

Require Not configured

Real-time protection ⓘ

Require Not configured

Previous Next

Fig 49 - Device security settings requiring Firewall and Antivirus

Included groups

⊞ Add groups ⊞ Add all users ⊞ Add all devices

Groups	Status	Group Members ⓘ	Filter	Filter mode	Edit filter
All devices	Active		None	None	Edit filter

Excluded groups

ⓘ When excluding groups, you cannot mix user and device groups across include and exclude. [Click here to learn more about excluding groups.](#)

+ Add groups

Groups	Status	Group Members ⓘ	Remove
No groups selected			

Fig 50 - Policy assigned to All devices

Why This Matters

A compliance policy defines the baseline security standards every managed device must meet. If a device falls below these standards it is marked non-compliant, which can then be used with Conditional Access to block the device from accessing company resources. This is how organisations enforce security across their entire device fleet from one central location.

Task 23 | Create an Intune Configuration Profile for Screen Lock

Microsoft Intune | Configuration Profiles | Device Lock

Created a Configuration Profile in Microsoft Intune called Screen Lock using the Settings Catalog. Configured device lock and inactivity timeout settings then assigned the profile to the 3rd Line Support group.

Steps

- In Intune Admin Centre went to Devices then Configuration and clicked Create Profile
- Selected Windows 10 and later as the platform and Settings catalog as the profile type

- Named the profile Screen Lock with description: Screen lock for users
- On the Configuration settings page clicked Add settings
- In the Settings picker searched for Lock screen and browsed to the Device Lock category
- Also searched for inactivity and selected Max Inactivity Time Device Lock from the Device Lock category
- Ticked both Max Inactivity Time Device Lock and Device Password Enabled
- Set Device Password Enabled to Enabled
- Set Max Inactivity Time Device Lock to 3 minutes
- Set Max Inactivity Time Device Lock With External Display to 3 minutes
- Clicked Next through Scope tags then on Assignments added the 3rd Line Support group
- Reviewed the settings on the Review and create page and clicked Create

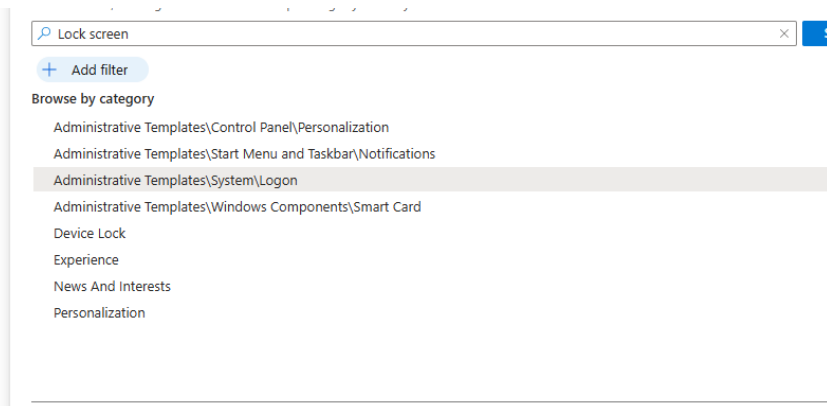


Fig 51 - Settings picker showing Lock screen categories available in the Settings catalog

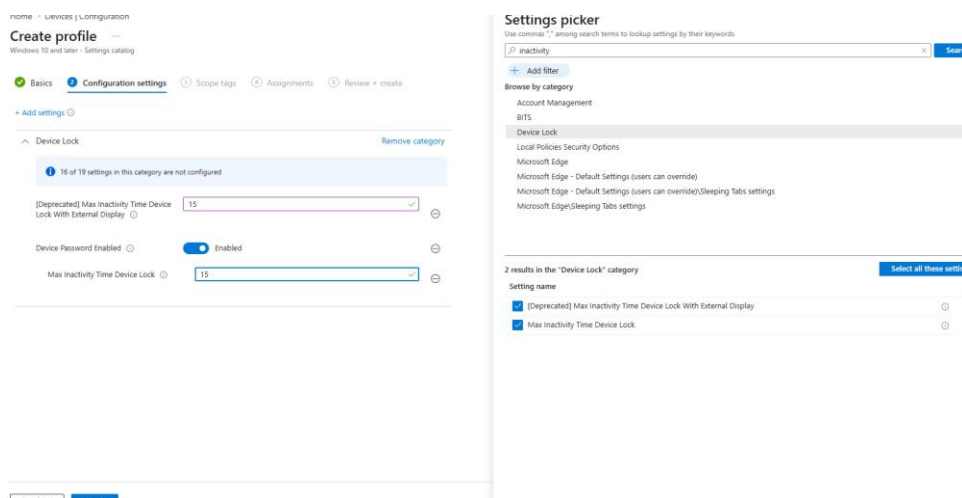


Fig 52 - Configuration settings showing Device Lock with inactivity timeout set to 3 minutes

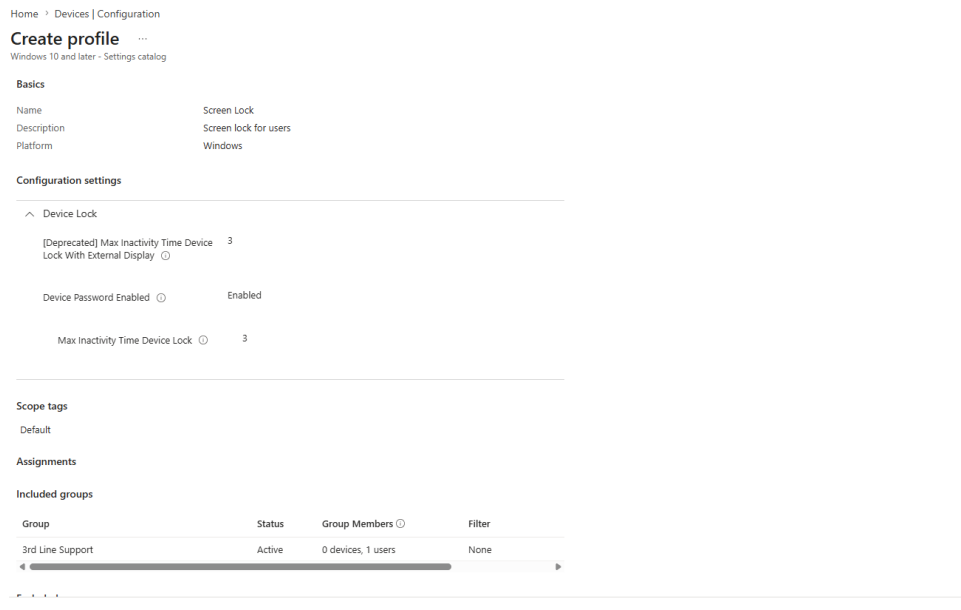


Fig 53 - Profile review page showing Screen Lock assigned to 3rd Line Support group

Why This Matters

Configuration profiles push settings directly to devices rather than just checking compliance. This profile enforces a screen lock after 3 minutes of inactivity on all devices used by 3rd Line Support staff. Unlike a GPO which requires on-premises Active Directory, Intune configuration profiles work for cloud-managed devices anywhere in the world, making them essential for hybrid and remote working environments.

Task 24 | Create a Password Policy Profile and Assign to a Security Group

Microsoft Entra ID | Microsoft Intune | Configuration Profiles

Created a Security Group in Entra ID for the Sales team, then built a Password Policy configuration profile in Intune with specific password requirements and assigned it directly to that group.

Step 1 - Create the Security Group in Entra ID

- Opened Entra ID in the Microsoft 365 Admin Centre and navigated to Groups
- Clicked New group and selected Security as the group type
- Named the group SG-PasswordPolicy-salesteam
- Added 2 Sales team users as members and clicked Create

Step 2 - Create the Password Policy Profile in Intune

- In Intune Admin Centre went to Devices then Configuration and clicked Create Profile
- Selected Windows 10 and later as the platform and Settings catalog as the profile type
- Named the profile Password Policy with description listing all the settings
- Clicked Add settings and searched for the Device Lock category
- Added and configured Maximum Password Age: 90 days
- Added and configured Minimum Password Length: 10 characters
- Added and configured Password Complexity: 1 (complexity required)
- Clicked Next through Scope tags

Step 3 - Assign the Profile to the Security Group

- On the Assignments page clicked Add groups under Included groups

- Searched for and selected SG-PasswordPolicy-salesteam
- Confirmed the group showed as Active with 0 devices and 2 users
- Left Excluded groups empty and clicked Next
- Reviewed all settings on the Review and create page and clicked Create

Home

Device Configuration

Create profile

Windows 10 and later - Settings catalog

✓ Basics

✓ Configuration settings

✓ Scope tags

✓ Assignments

Review + create

Summary

Basics

Name

Password Policy

Description

Password Policy
Minimum password length - 10
Password complexity - Required
Maximum password age - 90 days

Platform

Windows

Configuration settings

Device Lock

Maximum Password Age ⓘ

90

Minimum Password Length ⓘ

10

Password Complexity ⓘ

1

Scope tags

Default

Assignments

Included groups

Fig 54 - Password Policy profile review showing Device Lock settings before creation

Scope tags

Default

Assignments

Included groups

Group	Status	Group Members ⓘ	Filter
SG-PasswordPolicy-salesteam	Active	0 devices, 2 users	None

Excluded groups

Group	Status	Group Members ⓘ
No results.		

Fig 55 - Profile assigned to SG-PasswordPolicy-salesteam with 2 users active

Why This Matters

This demonstrates the correct enterprise workflow for targeted policy deployment. By creating a dedicated Entra ID security group for the Sales team and assigning the Intune profile to it, the password policy only applies to Sales staff rather than the whole organisation. This approach makes it easy to manage — adding a new Sales employee to the group automatically applies the policy to them, and removing them instantly revokes it.

Task 25 | Deploy an Application to Devices via Microsoft Intune

Microsoft Intune | App Deployment | Line-of-Business Apps

Deployed Google Chrome to the Sales team by uploading the enterprise MSI installer to Intune as a Windows line-of-business app and assigning it to the Sales team security group.

Steps

- Downloaded the Google Chrome Enterprise MSI installer (googlechromestandaloneenterprise64.msi)
- In Intune Admin Centre went to Apps then Windows and clicked Add
- Selected Windows MSI line-of-business app as the app type and clicked Select
- Uploaded the MSI file and waited for it to process
- Filled in app name: Google Chrome, Publisher: Google
- Added description: Google Chrome for only sales team
- Set App install context to Device and set Ignore app version to Yes
- Clicked Next and on the Assignments page assigned the app to the Sales team security group
- Reviewed all settings and clicked Create
- Confirmed the app appeared in Intune with status Install Pending confirming deployment is in progress

App Details

- **App name:** Google Chrome
- **Publisher:** Google
- **Package file:** googlechromestandaloneenterprise64.msi
- **Version:** 150.0.7871.47
- **Platform:** Windows
- **Assigned:** Yes

Deployment Status

- 1 device total, Install Pending: 1 — deployment queued and waiting for device sync

The screenshot shows the 'Add App' page in Microsoft Intune, specifically for a 'Windows MSI line-of-business app'. The 'App information' tab is active, showing various configuration fields. The 'Name' field is 'Google Chrome', 'Description' is 'Google Chrome for only sales team.', 'Publisher' is 'Google', 'App install context' is 'Device', 'Ignore app version' is 'Yes', and 'Show this as a featured app' is 'No'. The 'Next' button is highlighted at the bottom.

Fig 56 - App information page showing Google Chrome MSI being configured for deployment

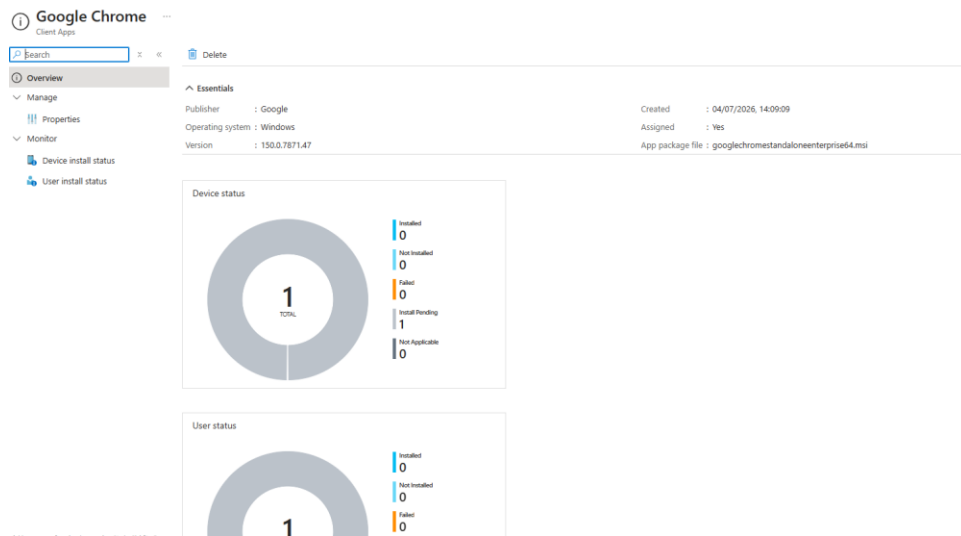


Fig 57 - Google Chrome app overview in Intune showing device status with install pending

Why This Matters

App deployment through Intune removes the need to manually install software on each device. An admin uploads the installer once and Intune pushes it automatically to every device in the assigned group. This is how IT teams manage software at scale across an organisation without touching individual machines. The Install Pending status confirms the policy has been pushed and will apply the next time the device checks in with Intune.

Task 26 | Create a User in Microsoft Entra ID and Add to a Group

Microsoft Entra ID | User Management | Group Assignment

Created a new user directly in Microsoft Entra ID and assigned them to the Finance group as part of the onboarding process.

Steps

- Opened the Microsoft Entra ID admin centre and navigated to Users
- Clicked New user and selected Create new user
- Set the User principal name to Tjames and selected the domain
- Set the Display name to James and Mail nickname to Tjames
- Set a password and confirmed Account enabled was set to Yes
- Clicked Next through Properties and moved to Assignments
- Under Groups clicked Add group and selected Finance
- Reviewed the settings and clicked Create to complete the user creation

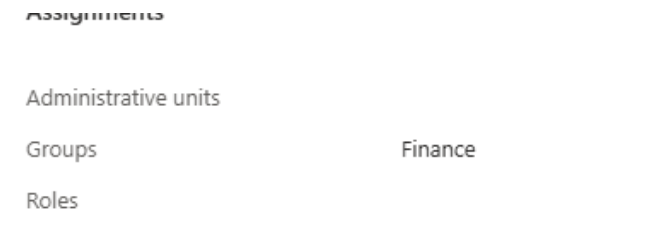
User Details

- **User principal name:** Tjames@hamzadahir2.onmicrosoft.com
- **Display name:** James
- **Mail nickname:** Tjames
- **Account enabled:** Yes
- **Group assigned:** Finance



Basics	
User principal name	Tjames@hamzadahir2.onmicrosoft.com
Display name	James
Mail nickname	Tjames
Password	*****
Account enabled	Yes

Fig 58 - New user James created in Microsoft Entra ID with account enabled



Assignments	
Administrative units	
Groups	Finance
Roles	

Fig 59 - User assigned to the Finance group during account creation

Why This Matters

Creating users directly in Entra ID is the cloud equivalent of creating users in on-premises Active Directory. Once created, James can sign in to any Microsoft 365 service including Outlook, Teams and SharePoint. Adding him to the Finance group at creation immediately gives him access to all shared resources that group controls, including any Intune policies, shared mailboxes or SharePoint sites assigned to Finance.

Task 27 | Create Security Groups with Assigned Entra Roles and Add Users to Groups

Microsoft Entra ID | Security Groups | Role-Based Access Control

Created security groups in Microsoft Entra ID and assigned an admin role directly to a group, then created a user and added them to the security group they belong to so they inherit the role through membership rather than being assigned it individually.

Step 1 - Create the Security Group and Assign a Role

- Opened the Microsoft Entra ID admin centre and navigated to Groups
- Clicked New group and selected Security as the group type
- Named the group SG-2ndLine-Support and gave it a matching description
- Set Microsoft Entra roles can be assigned to the group to Yes
- Left Membership type set to Assigned
- Assigned the User Administrator role to the group and clicked Create

Step 2 - Create the User

- In Microsoft Entra ID clicked New user and selected Create new user
- Set the User principal name to Sjohn@hamzadahir2.onmicrosoft.com
- Set Display name to John Smith and Mail nickname to Sjohn
- Set a password and confirmed Account enabled was set to Yes
- Clicked Next through Properties to reach Assignments

Step 3 - Add the User to Their Security Group

- On the Assignments page clicked Add group under Groups
- Searched for and selected SG-1stLine-Support and Tech Support Team
- Reviewed all settings on the Review + create page and clicked Create to complete the user creation

Group Details

- **Group type:** Security
- **Group name:** SG-2ndLine-Support
- **Microsoft Entra roles assignable:** Yes
- **Membership type:** Assigned
- **Role assigned:** User Administrator

The screenshot shows the 'Review + create' page for a new Microsoft Entra security group. At the top, there is a 'Got feedback?' link. The form contains the following fields and settings:

- Group type:** A dropdown menu set to 'Security'.
- Group name:** A text box containing 'SG-2ndLine-Support,' with a green checkmark icon on the right.
- Group description:** A text box containing 'SG-2ndLine-Support,' with a green checkmark icon on the right.
- Microsoft Entra roles can be assigned to the group:** Two radio buttons, 'Yes' (selected) and 'No'.
- Membership type:** A dropdown menu set to 'Assigned'.
- Owners:** A section with the text 'No owners selected'.
- Members:** A section with the text 'No members selected'.
- Roles:** A section with the text 'User Administrator'.

Fig 60 - SG-2ndLine-Support security group created with Microsoft Entra roles enabled and the User Administrator role assigned

User Details

- **User principal name:** Sjohn@hamzadahir2.onmicrosoft.com
- **Display name:** John Smith
- **Mail nickname:** Sjohn
- **Account enabled:** Yes
- **Groups assigned:** SG-1stLine-Support, Tech Support Team

Create new user

Create a new internal user in your organization

Basics

Properties

Assignments

Review + create

Basics

User principal name	Sjohn@hamzadahir2.onmicrosoft.com
Display name	John Smith
Mail nickname	Sjohn
Password	
Account enabled	Yes

Properties

First name	John
Last name	Smith
User type	Member

Assignments

Administrative units	
Groups	SG-1stLine-Support Tech Support Team
Roles	

Fig 61 - New user John Smith created and assigned to the SG-1stLine-Support and Tech Support Team groups

Why This Matters

Task 28 | Create a Conditional Access Policy to Restrict Access by Location

Created a named location for the UK and built a Conditional Access policy that blocks access to company resources from anywhere except that location, so only users connecting from the UK can sign in.

Step 1 - Create the Named Location

Name	Location type	Trusted	Conditional Access policies	Creation date	Modified date
UK Only	Countries (IP)		Not configured in any policy yet	09/07/2026, 13:51:15	09/07/2026, 13:51:15

Fig 62 - UK Only named location created, using Countries (IP) as the location type

Step 2 - Create the Conditional Access Policy

- In Conditional Access clicked New policy and named it Block Access Outside UK
- Under Users or agents included All users and excluded one admin account so it could not be locked out by the policy
- Under Target resources selected All resources (formerly All cloud apps)
- Under Conditions opened Network, set Configure to Yes, and on the Include tab initially selected UK Only
- Realised this would block the UK rather than restrict access to it, so switched to the Exclude tab, selected Selected networks and locations, and chose UK Only there instead
- The Network condition summary then read Any network or location and 1 excluded, meaning every location is blocked except the UK
- Under Access controls, Grant, selected Block access
- Left Enable policy set to Report-only to test the policy's impact before enforcing it, then clicked Save

Policy Details

- **Policy name:** Block Access Outside UK
- **Users or agents:** All users included, 1 user excluded
- **Target resources:** All resources
- **Network condition:** Any network or location, excluding UK Only
- **Grant control:** Block access
- **State:** Report-only

Control access based on Conditional Access policy to bring signals together, to make decisions, and enforce organizational policies. [Learn more](#)

Name *

Block Access Outside UK

Assignments

Users or agents ⓘ

All users included and specific users excluded

Target resources ⓘ

All resources (formerly All cloud apps)

Network ⓘ

Any network or location and 1 excluded

Conditions ⓘ

1 condition selected

Access controls

Enable policy

Report-only On Off

⚠ It looks like you're about to manage your organization's security configurations. That's great! You must first [disable security defaults](#) before enabling a...

Save

Control user access based on their network or physical location. [Learn more](#)

Configure ⓘ

Yes No

Include Exclude

Select the locations to exempt from the policy

☐ All trusted networks and locations

☐ All Compliant Network locations

☒ Selected networks and locations

Select

UK Only

UK Only ***

ⓘ To create a Conditional Access policy ensuring your tenant's members are coming from their compliant network, make sure Global Secure Access (GSA) is

Fig 63 - Network condition corrected to exclude UK Only, so every location is blocked except the UK

Why This Matters

Conditional Access lets an organisation enforce access rules based on real-time signals such as location, device compliance or user risk, without needing a VPN or on-premises firewall rule. Including UK Only under Exclude rather than Include is the detail that makes the policy work as intended, since Include would have blocked the UK instead of restricting access to it. Keeping the policy in Report-only mode first lets an admin see who the policy would affect before it actually blocks anyone, and excluding an admin account is a standard safeguard against getting locked out while testing.